Audience Effect in Microfinance:

Does Social Observability of Default Increase Repayment?

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Abstract

Many microfinance institutions have progressively individualized lending, first through a shift from joint to individual liability with group based collection practices, and recently to individual collection practices as a potential next step. This paper asks whether the latter individualization would affect repayment by eliminating group practices that make default observable to the community. In a framed lab-in-the-field experiment, I vary the *social observability of default* within subject as borrowers make repayment decisions on a hypothetical loan under stochastic income, where higher repayment reduces their monetary payoff. This variation identifies the *audience effect* on repayment. I find that an audience raises repayment by 3.3 to 5.3 percent relative to the amount repaid without an audience. Repayment without an audience rises with income, ranging from half to two-thirds of the debt obligation, while the audience effect does not vary detectably across income states. The conceptual framework interprets repayment without an audience as reflecting intrinsic motivation and the additional repayment induced by an audience as reflecting reputational motivation. Although the estimated repayment gain is modest relative to the total debt obligation, an equivalent reduction in field default would raise lender profit by 34 to 55 percent of the zero-default margin.

Keywords: microfinance, individual liability, lab-in-the-field experiment, social observability, audience effect, behavioral motivation, reputation

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1 Introduction

Microfinance expanded credit access to low-income households by substituting physical collateral with social and relational enforcement mechanisms. Early microfinance institutions relied on joint liability contracts and group-based collection practices to enforce repayment. In joint liability contracts, shared financial responsibility creates pecuniary incentives for assortative matching, peer monitoring, and social sanctions (Ghatak, 2001; Stiglitz, 1990; Besley and Coate, 1995). Although many lenders have later transitioned to individual liability due to concerns about hidden costs such as free riding, collusion, and coordination burdens, as well as increased competition, studies of this transition found no reduction in repayment rates (Madajewicz, 2011; Giné et al., 2010). While this shift removed collective financial responsibility, lenders retained group collection practices (Rai and Sjöström, 2010; De Quidt et al., 2018). Institutions such as ASA, BRAC, and Grameen Bank II continue to organize borrowers into groups from the same neighborhoods and collect repayments at group meetings (Armendáriz and Morduch, 2010). More recently, lenders are exploring further individualization of loan collection practices, as digital payments have made individual loan transactions increasingly feasible and cost-effective (Flaming et al., 2023; BRAC International, 2022; Heath and Riley, 2024). This next stage of individualization would remove group collection and follow-up practices that make delinquency observable to peers and the wider community. While pecuniary incentives such as credit termination and dynamic incentives remain, it is unclear how costly it would be to eliminate *social observability of default*, a remaining social enforcement mechanism in individual liability microfinance.

In this paper, I examine whether eliminating social observability of default affects repayment in microfinance. Identifying this effect empirically has been difficult as observability is often an implicit feature of such lending environment and provides little empirical variation. Moreover, individual liability contracts retain pecuniary incentives such as termination threats and progressive lending to sustain repayment.[1] The behavioral economics literature emphasizes intrinsic and reputational motivations as sources of nonpecuniary incentives for individual behavior (Bénabou and Tirole, 2006). Akerlof (1980) shows that deviation from social norms generates disutility through community disapproval and, for some individuals, through the violation of internalized values. Andreoni and Bernheim (2009) similarly show that individuals may care not only about behaving fairly but also about being perceived as fair by others, which they describe as an *audience effect*. Adapted to the microfinance context, these mechanisms imply that repayment may be sustained by a self-imposed obligation to honor the loan, while lender-imposed social observability of default may add concerns about reputational loss or being perceived as violating established norms. After stripping away other pecuniary mechanisms,

[1]Termination incentives and progressive lending have been studied theoretically and empirically by Armendáriz and Morduch (2000), Giné et al. (2010), and Giné et al. (2012). Frequent repayment schedules have also been viewed as a means of cultivating financial discipline, although Field and Pande (2008); McIntosh (2008) find little difference in default rates when installment frequency varies. This paper focuses on behavioral motivations in the absence of these pecuniary incentives.

the experiment distinguishes repayment when default is not socially observable from the additional repayment induced by observability, which the conceptual framework interprets as reflecting intrinsic and reputational motivations.

To study the behavioral motivations, I conduct a framed lab-in-the-field experiment with 919 active BRAC microfinance borrowers across 100 villages in southern Bangladesh. I vary the social observability of default within subjects as borrowers facing stochastic income make repayment decisions on a hypothetical loan. Borrowers are presented with a realistic rice-farming scenario involving three income states good, normal, and bad with a fixed debt obligation. Greater default leaves borrowers with more experimental income, which is converted into cash payments. In the private game (P), borrowers choose how much to repay knowing that their default amounts will remain confidential. Repayment therefore involves a tradeoff between the monetary gains from default and their intrinsic motivation to repay. In the audience game (A), borrowers make the same decision but are informed that their default amounts will be disclosed to peers and community members, similar to how delinquency becomes known in actual lending settings. To make this disclosure explicit, we display a board listing the names of participating group members, with space next to each name to record the default amount. Thus, borrowers face an additional reputational consideration in the audience game, along with the monetary and intrinsic motivations present in the private game. The difference in repayment between the two conditions identifies the audience effect. Sessions were conducted at village meeting centers with groups of 6 to 10 borrowers from existing microfinance groups, resembling the setting in which actual loan transactions occur. The order of the private and audience games, as well as the income states, was randomized at the session level.

I find that an audience increases repayment by 3.3 to 5.3 percent relative to repayment without an audience. The OLS fixed effects estimates show that an audience raises observed repayment by 3.3–3.5 percent across income states. These estimates may understate the underlying latent repayment response because repayment is censored at the lower and upper bounds of the debt obligation, with up to forty percent of observations at a boundary. I therefore complement them with a two-limit Tobit specification that accounts for censoring and estimates a latent audience effect of 3.8–5.3 percent. A Monte Carlo exercise calibrated to the experimental design shows that censoring attenuates the observed effect relative to the latent effect, while the Tobit estimator recovers the latent effect with little bias. The estimates further show that repayment is state dependent, increasing with realized income. Without an audience, borrowers repay 34 percent more in the good state than in the adverse state. Although full repayment is feasible in all income states and would leave borrowers with no game earnings in the adverse state, repayment without an audience is about one-half to two-thirds of the debt obligation across states. Interestingly, the audience effect does not vary detectably across income states in absolute terms, indicating that social observability adds repayment incentives to the state-dependent intrinsic

motivation rather than substituting for it.

Since repayment norms may vary across borrowers, the within-subject design controls for time invariant individual heterogeneity and can improve the precision of the estimated audience effect (List et al., 2011; Charness et al., 2012). In the data, borrower fixed effects increase the explained variation in repayment from 2–3 percent to 66–73 percent across income states. However, repeated decisions raise two concerns. First, whether repeated choices reflect systematic borrower behavior rather than idiosyncratic responses. Repayment choices are consistent within borrowers across income states and rounds, with reliability coefficients ranging from 0.81 to 0.87, well above those expected under random choice. Second, whether the audience effect changes over the course of the session or depends on the order in which the conditions are presented. Although repayment declines across rounds, the audience effect does not vary detectably by round, and randomized game order reveals no detectable order asymmetry. Round fixed effects absorb the systematic decline in repayment, while the audience effect is identified from within-borrower differences between private and audience conditions. I report the pooled coefficient controlling for the round fixed effects as the average treatment effect.

Although modest relative to the total debt obligation, the repayment gain can be economically important for lender profitability (Cull et al., 2009; Rosenberg et al., 2013; Flatnes and Carter, 2019). When lending margins are thin, a small reduction in default can generate a large proportional increase in profit. The estimated increase in observed expected repayment of 675.5 Taka corresponds to a 1.93 percentage-point reduction in the default share of the total debt obligation. If a default reduction of the same magnitude occurred on an otherwise comparable field loan, it would raise lender profit by 34 to 55 percent of the zero-default margin.[2]

The paper makes three contributions. First, it contributes to the microfinance literature by estimating the effect of an implicit contract feature in individual liability. The literature on microfinance enforcement has focused on joint liability contracts, where shared financial responsibility creates incentives for peer monitoring, social sanctions, and assortative matching (Stiglitz, 1990; Besley and Coate, 1995; Ghatak, 2001; Armendáriz and Morduch, 2000; Giné et al., 2012; Balboni et al., 2022). Yet evidence from individual liability lending shows that mechanisms other than joint liability can also sustain repayment. Giné and Karlan (2014) find no increase in default when joint liability is removed. In laboratory games with microentrepreneurs in Lima, Giné et al. (2010) find that dynamic incentives reduce risky project choice and default across loan contracts, including individual liability. Other studies show that group meetings can strengthen social interaction and risk sharing (Feigenberg et al., 2013; De Quidt et al., 2016). Two studies are closely related to this paper in their focus on behavioral mechanisms in individual liability microfinance. Dhami et al. (2022) examine how shame, made salient by communicating social

[2]The accounting exercise does not account for additional costs of implementing social observability. Such costs may be limited where lenders already collect repayments through village meetings or household visits, but observability could still require additional staff time or affect borrower retention and group relationships.

norms and allowing peers to express approval or disapproval, affects borrowers' ex-ante effort choices in experimentally assigned groups. I instead examine ex-post repayment decisions among borrowers in existing microfinance groups, where default is disclosed to peers without explicitly eliciting approval or disapproval. The audience effect reflects the combined response to lender-imposed social observability rather than shame alone. In a field experiment, Bu and Liao (2022) find that threatening public disclosure of delinquency reduces the share of overdue installments, although observed delinquency may reflect both borrowers' ability and willingness to repay. I complement their findings by studying ex-post moral hazard among borrowers whose selection into borrowing and group formation occurs before the experiment. By experimentally varying income states and whether default is observable to peers, I identify the audience effect on willingness to repay at a given income level, excluding lender observability and contractual monetary incentives.

Second, the paper brings behavioral theories of norm driven decision making to the study of repayment incentives in a formal credit market. Experimental literature uses social preference games show that people act more generously when their choices are observed (Kimbrough and Vostroknutov, 2016; Andreoni and Bernheim, 2009), and that norm-conforming behavior depends on how choices can be seen and justified (Dana et al., 2007). Related evidence from credit markets shows that moral appeals can improve repayment (Bursztyn et al., 2019), while Czura et al. (2026) find high repayment even when repayment is not monetarily optimal, consistent with intrinsic repayment norms. Evidence from savings decisions further shows that making financial behavior observable to others can strengthen reputational incentives (Breza and Chandrasekhar, 2019). I extend these insights to an incentivized repayment decision framed around a formal credit environment, where greater default increases monetary payoffs. The experiment distinguishes repayment when default remains private from the additional response induced when default becomes observable to peers. I find that private repayment varies with the borrower's realized income, while the audience effect does not vary detectably across income states. These findings suggest that borrowers adjust their private repayment decisions to economic circumstances, while social observability provides an additional repayment incentive across those circumstances.

Third, the paper provides a within-subject experimental design for characterizing borrower-specific repayment behavior. Repeated repayment decisions across private and audience conditions and different income states allow measuring repayment without an audience and the additional repayment response to social observability at the borrower level. In the conceptual framework, I interpret these measures as reduced-form indicators of intrinsic and reputational motivations. These measures provide a basis for future research examining whether individual differences in repayment behavior predict field outcomes, such as borrowing decisions and demand for risk-management products, building on work linking experimental behavior to real-world economic choices (Castillo et al., 2018).

The rest of the paper proceeds as follows. Section 2 presents a conceptual framework, that shows how behavioral motivations enter borrowers repayment decisions. Section 3 describes the experimental design, and Section 4 outlines the estimands and empirical strategy. Section 5 reports the main results, Section 6 examines heterogeneity, Section 7 discusses the implications for lender profitability, and Section 8 concludes with interpretation and policy implications.

2 Conceptual Framework

The conceptual framework adapts the characterization of behavioral motivations in Bénabou and Tirole (2006) to the institutional setting of individual liability microfinance. While they model prosocial behavior as a combination of intrinsic, extrinsic, and reputational motivations, I simplify the model to focus on intrinsic and reputational motivations. I model repayment as a tradeoff between the monetary gains from nonrepayment and the disutility of deviating from repayment norms, with borrowers differing in the value they place on these motivations.

2.1 A Model of Behavioral Motivations for Repayment

Let borrower i choose repayment amount $r \in [0, \bar{r}]$ after observing her income $y(\theta_i)$, where $\bar{r}$ is the fixed debt obligation and $\theta_i \in [\underline{\theta}, \bar{\theta}]$ indexes the income realization, with $y'(\theta_i) > 0$. Let $s \in \{0, 1\}$ denote whether the borrower's default is socially observable, or equivalently, whether there is an audience for r. The borrower's utility consists of three components,

$$U(r; s, \theta_i, \mathbf{x}_i) = \underbrace{u(y(\theta_i) - r)}_{\substack{\text{utility from} \\ \text{retained earning}}} + \underbrace{\alpha_i \, N\Big(r - n^I(\theta_i)\Big)}_{\substack{\text{disutility from violating} \\ \text{the intrinsic norm}}} + \underbrace{s \, \gamma_i \, N\Big(r - n^R(\mathbf{x}_i)\Big)}_{\substack{\text{disutility from violating} \\ \text{the reputational norm}}} \tag{1}$$

with $u'(\cdot) > 0$ and $u''(\cdot) < 0$. The disutility from violating the intrinsically motivated norm depends on $n^I(\theta_i)$, the borrower's intrinsic belief about how much she ought to repay in state θ_i, and on $\alpha_i \geq 0$, the value she places on adhering to her intrinsic value. This intrinsic motivation affects repayment regardless of whether default is observed by others. The disutility from violating the reputationally motivated norm depends on $n^R(\mathbf{x}_i)$, what the borrower believes others expect her to repay given her observable characteristics $\mathbf{x}_i$, and on γ_i, the value she places on adhering to the reputational expectation. This reputational motivation enters only when $s = 1$, when default is socially observable. The penalty function is common to both norms, $N(z) = \widetilde{N}(\min\{z, 0\})$, with $N'(z) \geq 0$ and $N''(z) \leq 0$ for $z < 0$. Only repayment below the relevant norm generates disutility, while repayment at or above the norm generates no additional utility. The utility function collapses to the two conditions,

$$(s=0): \quad U = u(y(\theta_i) - r) + \alpha_i N(r - n^I(\theta_i))$$

$$(s=1): \quad U = u(y(\theta_i) - r) + \alpha_i N(r - n^I(\theta_i)) + \gamma_i N(r - n^R(\mathbf{x}_i)).$$

Without social observability, optimal repayment $r^*(0)$ reflects the trade off between the monetary gain from retaining income and the disutility from repaying below the intrinsically motivated norm. With social observability, optimal repayment $r^*(1)$ reflects the same trade off and the additional disutility from repaying below the reputationally motivated norm. The repayment gain from social observability is $\Delta_i \equiv r^*(1) - r^*(0)$, which I refer to as the *audience effect* in the reduced form analysis.

The conceptual framework does not explicitly model posterior inference by peers. In a richer signaling framework, peers would observe repayment, form a judgment about the borrower, and the borrower would anticipate that judgment when choosing how much to repay. I instead summarize the repayment norm she believes others expect her to meet through $n^R(\mathbf{x}_i)$ and the value she places on adhering to that norm through γ_i. Borrowers may similarly differ in the intrinsic norm $n^I(\theta_i)$ and in the value they place on adhering to it, α_i. Thus, intrinsic and reputational motivations may differ across borrowers because of differences in the relevant repayment norm, the value placed on adhering to it, or both. The reduced-form analysis therefore estimates the composite effect of these motivations on repayment rather than separately identifying their underlying components.

2.2 Comparative Statics

Social Observability. Assume $u'(\cdot) > 0$, $u''(\cdot) < 0$, $N'(\cdot) > 0$, and $N''(\cdot) \leq 0$ over the relevant range, so that utility is concave in repayment and the second-order condition $u''(\cdot) + \alpha_i N''(\cdot) < 0$ holds. The first-order conditions for an interior optimum are,

$$-u'(y(\theta_i) - r^*(0)) + \alpha_i N'(r^*(0) - n^I(\theta_i)) = 0 \tag{2}$$

$$-u'(y(\theta_i) - r^*(1)) + \alpha_i N'(r^*(1) - n^I(\theta_i)) + \gamma_i N'(r^*(1) - n^R(\mathbf{x}_i)) = 0 \tag{3}$$

In both cases, the borrower equates the marginal monetary cost of repayment with the marginal disutility from norm violation. With social observability, an additional marginal cost arises from violating the reputational repayment norm through $\gamma_i N'(r^*(1) - n^R(\mathbf{x}_i))$.

Proposition 1 (Effect of social observability on repayment)**.** *Let $r^*(0)$ and $r^*(1)$ denote the optimal repayment levels without and with social observability, respectively. If $\gamma_i > 0$ and $N'(r^*(1) - n^R(\mathbf{x}_i)) > 0$ at an interior optimum, then introducing an audience strictly increases optimal repayment, so that $r^*(1) > r^*(0)$. Equality holds only at a corner solution or where $N'(\cdot) = 0$. The individual repayment gain $\Delta_i \equiv r^*(1) - r^*(0)$ is weakly increasing in γ_i, and the average audience effect is $\mathbb{E}[\Delta_i]$.*

Income State. Applying the implicit function theorem to equations (2) and (3) gives,

$$\frac{\partial r^*(0)}{\partial \theta_i} = \frac{u''(y(\theta_i) - r^*(0))y'(\theta_i) + \alpha_i N''(r^*(0) - n^I(\theta_i))\dfrac{dn^I(\theta_i)}{d\theta_i}}{u''(y(\theta_i) - r^*(0)) + \alpha_i N''(r^*(0) - n^I(\theta_i))}. \tag{4}$$

$$\frac{\partial r^*(1)}{\partial \theta_i} = \frac{u''(y(\theta_i) - r^*(1))y'(\theta_i) + \alpha_i N''(r^*(1) - n^I(\theta_i))\dfrac{dn^I(\theta_i)}{d\theta_i}}{u''(y(\theta_i) - r^*(1)) + \alpha_i N''(r^*(1) - n^I(\theta_i)) + \gamma_i N''(r^*(1) - n^R(\mathbf{x}_i))}. \tag{5}$$

Without socially observable default in equation (4), the denominator is negative under $u'' < 0$ and $N'' \leq 0$. The numerator contains two components. The first, $u''(y(\theta_i) - r^*(0))y'(\theta_i) < 0$ when $y'(\theta_i) > 0$, reflects an income effect. A better income state raises available resources, lowers the marginal utility of retained income, and therefore lowers the opportunity cost of repayment. The second term is non-positive when $dn^I(\theta_i)/d\theta_i \geq 0$, reflecting an intrinsic norm response. If the borrower's intrinsic repayment norm rises with the income state, deviating from that norm becomes more costly, further increasing the marginal utility of repayment. Thus, the numerator is non-positive and $\partial r^*(0)/\partial \theta_i \geq 0$.

The same two components determine state dependence when default is socially observable. The denominator of equation (5) remains negative and additionally reflects the curvature of the disutility from violating the reputational norm. Thus, $\partial r^*(1)/\partial \theta_i \geq 0$ under the same conditions. For a constant marginal disutility for norm violations and interior repayment choices, the audience effect Δ_i is independent of the income state. The reputational norm $n^R(\mathbf{x}_i)$ depends only on borrower characteristics observable to the audience, so unobserved characteristics, such as idiosyncratic income realizations, do not enter the norm, $\partial n^R(\mathbf{x}_i)/\partial \theta_i = 0$. While borrowers repay less in worse income states because adverse states raise the marginal value of retained income against a lower cost of violating the repayment norms, the reputational norm therefore affects the magnitude of the repayment response through the denominator without introducing a separate state-dependent term in the numerator.

Proposition 2 (State dependence of repayment)**.** *Suppose $u'' < 0$, $N'' \leq 0$, $y'(\theta_i) > 0$, and $dn^I(\theta_i)/d\theta_i \geq 0$. Then optimal repayment is weakly increasing in θ_i both without and with social observability, so that $\partial r^*(0)/\partial \theta_i \geq 0$ and $\partial r^*(1)/\partial \theta_i \geq 0$. With constant marginal disutility from norm violations and interior repayment choices, the audience effect is independent of the income state, $\partial \Delta_i/\partial \theta_i = 0$.*

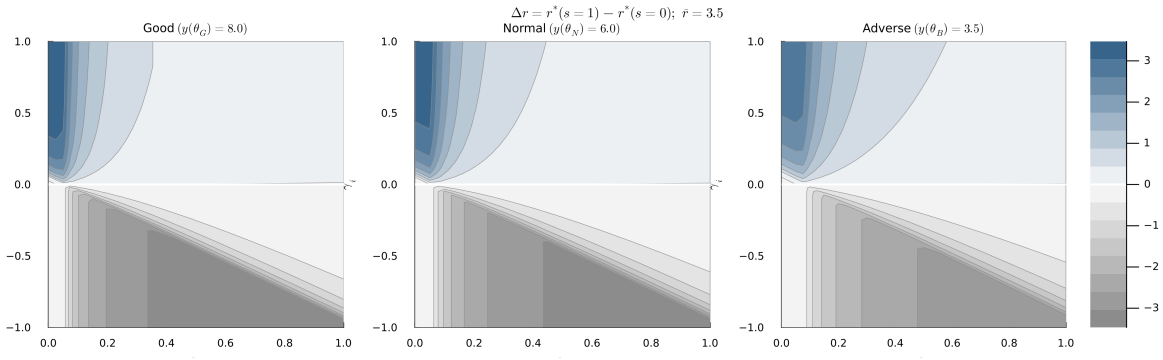
2.3 Repayment Motivation and Norm Heterogeneity

The comparative statics examine the effect of social observability holding socioeconomic characteristics fixed. Borrower characteristics may nevertheless be associated with both private repayment and the audience effect through differences in behavioral parameters. First, consider individual specific differences in the value one places on adhering to norms. Kimbrough and Vostroknutov (2016) show that individuals differ in their sensitivity to norm violations and willingness to incur monetary costs to

comply with norms. In microfinance, borrowers may differ in α_i and γ_i due to upbringing, socialization, and past experience. When norms favor higher repayment, an increase in $\alpha_i > 0$ raises repayment even when default remains private, whereas social observability increases repayment only if borrowers place positive value on adhering to the reputational norm ($\gamma_i > 0$). Those who rely on informal insurance may place greater value on maintaining a favorable reputation within their community (Udry, 1994). Moreover, Better off borrowers may value status signaling as a way to maintain their social standing in the community (Bricker et al., 2021).

However, social observability need not always encourage higher repayment, as lower repayment may also be reputationally valuable for some borrowers, such that $\gamma_i < 0$. Bursztyn and Jensen (2015) show that observability can have opposite effects depending on what borrowers believe it signals to peers. This may depend on the nature of their social ties with the audience (Breza and Chandrasekhar, 2019). An individual may prefer to conceal financial solvency if doing otherwise invites demands for assistance and social obligations (Baland et al., 2011). Lower repayment could also signal bargaining strength, financial prudence, prioritization of household needs, or conformity with a local norm. Appendix B uses explicit functional forms to illustrate how the audience effect can be positive or negative over the $(\alpha_i \geq 0, \gamma_i \in \mathbb{R})$ parameter space.

Figure 2.1: Audience effect in α_i and γ_i space



Along with the value borrowers place on adhering to or violating norms, they may also differ in what they believe constitutes repayment norms. The intrinsic norm $n^I(\theta_i)$ reflects what borrowers believe they ought to repay given their circumstances, regardless of whether others observe their decisions. While the comparative statics show that an adverse income state raises the marginal disutility of repayment, such circumstances may also determine the intrinsic norm by providing a self-justification for lower repayment, a form of "moral wiggle room" examined by Dana et al. (2007). Some borrowers may regard full repayment as an unconditional moral obligation, while others may consider retaining income for essential household needs appropriate. Education may, for example, determine how borrowers assess these needs against the moral obligation to repay.

The reputational norm $n^R(\mathbf{x}_i)$ reflects what borrowers believe others expect them to repay given

information observable to the audience. Peers may expect higher repayment from borrowers with greater wealth or a history of reliable repayment, insofar as these characteristics signal repayment capacity or reliability. Exposure to a common shock, such as flooding, may instead lower repayment expectations if peers regard widespread hardship as a justification for lower repayment, even when a borrower's individual income is unaffected. Since peers may observe the common shock without knowing the borrower's actual income loss, the shock may change their repayment expectations without revealing individual repayment capacity.

The value borrowers place on adhering to norms and their belief in repayment norms may influence repayment in competing ways and offset one another. The empirical analysis identifies their combined effect through Δ_i, rather than separately identifying the underlying behavioral parameters and repayment norms. I interpret the heterogeneity results as reduced-form evidence on how the effect of social observability varies across borrowers.

3 Experimental Procedures and Data

This section describes the institutional setting and how the behavioral motivations in the conceptual framework map to the experimental conditions. It then presents the experimental design and procedures, data collection, and sample characteristics.

3.1 Institutional Setting and Mapping to Experiment

BRAC, one of the world's largest microfinance providers, operates individual liability lending in Bangladesh. Although borrowers are individually responsible for their loans, borrowing groups are formed in the village neighborhood. Borrowers meet monthly at village centers, where loan officers collect installments, or loan officers visit them at their homes if they fail to repay. If borrowers are absent, loan officers contact group leaders or neighbors to locate them and follow up on missed payments. Household visits also draw the attention of other community members to delinquency. In these close knit communities, such collection practices can make default observable to peers and other members of the local community, inducing reputational motivation for repayment.

Group based collection allows borrowers to repay loans within their neighborhoods, but maintaining these meetings and household visits requires loan officers' time and travel. With the expansion of mobile money, individual digital repayment has become increasingly feasible (Flaming et al., 2023). In Bangladesh, BRAC piloted loan repayments through bKash among small-business borrowers in 2011 (Hanouch and Rotman, 2013) and recently, an industry estimate reported by Jahan (2026) suggests that about 10% of microfinance transactions in Bangladesh are digital. BRAC has also piloted digital repayments among group loan clients in Tanzania, although group meetings were retained (Heath and

Riley, 2024). If digital repayment replaces group meetings and follow-up visits, it may reduce collection costs but also remove practices that make default socially observable. While contractual incentives, including loan termination and future access to credit, can remain unchanged, it is unclear whether eliminating these practices would weaken repayment by removing this remaining social enforcement mechanism.

Table 3.1 summarizes the mapping between the behavioral motivations in the conceptual framework and the experimental design. Since the experiment removes other contractual penalties, repayment when default remains private provides a measure of repayment associated with intrinsic motivation. The additional repayment induced when default becomes observable to peers identifies the audience effect, which I interpret as a reputationally motivated response. These repayments represent the conceptual framework's interpretation of the reduced-form responses rather than separately identified behavioral mechanisms.

Table 3.1: Mapping behavioral motivations to microfinance context and experimental conditions.

Behavioral Motivation[a]	**Microfinance Context**	**Experimental Condition**
Intrinsic: Motivation to undertake an action because it is valued in itself or is consistent with an internalized self-standard	Borrower's internalized sense of obligation to honor a loan	*Private Game (P):* Borrowers privately decide how much to repay and are reminded only of the debt obligation.
Reputational: Motivation to undertake an action because of how it would be perceived or evaluated by others	Borrower's motivation to maintain her social standing or reputation	*Audience Game (A):* Borrowers are informed that their default will be revealed to peers.

[a] The definitions draw on Andreoni and Bernheim (2009), Akerlof (1980), Bénabou and Tirole (2006), Cialdini et al. (1991), and Cialdini and Goldstein (2004). The experiment varies the social observability of default.

3.2 Experimental Design

The experiment presents a rice farming scenario in which each game represents a farming season and borrowers make repayment decisions under good, normal, and bad income states (Figure A.4). The hypothetical loan obligation is calibrated to borrowing for farming in the study population, while monthly installments are collapsed into a single end-of-season repayment decision to simplify the experimental task. Decisions are incentivized with real monetary payments. Sessions were conducted at village meeting centers with borrowers belonging to the same existing microfinance groups, resembling the setting in which actual loan transactions occur (Figure A.3).

In each game, borrowers face a fixed debt obligation of 35,000 Taka and choose repayment amount in increments of 5,000 Taka. Income is 80,000 Taka in the good state, 60,000 Taka in the normal state, and 35,000 Taka in the bad state, with corresponding probabilities of 0.30, 0.40, and 0.30. Borrowers

therefore have sufficient income to repay the full debt in all three states, although in the bad state, full repayment leaves them with no remaining income from the game, as can happen in real life.

In the private games, repayment and default amount remain confidential. Participants are reminded of their debt obligation and that they may retain the residual income after repayment, which increases with the amount defaulted. In the audience games, participants are informed that their default amount will be revealed to others present at the session, including members of their borrowing group, but not the income state in which that default amount was chosen. To make social observability salient, a board listing all participants' names is displayed with space to record individual default amounts (Figure A.5). Participants are informed that one of the audience games decision will be drawn at random for revelation after all repayment decisions are complete. In practice, no default amounts were revealed at the end of the session. Participants therefore made their decisions under the expectation of revelation, while actual disclosure was withheld to avoid extending any potential reputational consequences beyond the experiment.

I use a strategy method in which each borrower makes repayment choices for all combinations of game type, round, and state of the world before payoff determination. Each round contains one private and one audience game, with three income state decisions in each game. Each participant therefore makes $2 \times 4 \times 3 = 24$ repayment decisions across four rounds, including the practice round. The first round is for practice, leaving 18 incentivized decisions. To mitigate anchoring and order effects, the sequence of private and audience games was randomized at the session level, following one of the two sequences shown in Table 3.2. The order of the three income states and the presentation of contextual examples read by facilitators were also randomized at the session level.

Table 3.2: Experimental design.

		Practice		**Actual Game**					
		Round 1		Round 2		Round 3		Round 4	
		G1	G2	G3	G4	G5	G6	G7	G8
Panel A. Game condition (order randomized)									
Sequence 1		**P**	**A**	**A**	**P**	**P**	**A**	**A**	**P**
Sequence 2		**A**	**P**	**P**	**A**	**A**	**P**	**P**	**A**
Panel B. State of the world within each game (order randomized)									
Good	80,000 ($p = .30$)	•	•	•	•	•	•	•	•
Normal	60,000 ($p = .40$)	•	•	•	•	•	•	•	•
Bad	35,000 ($p = .30$)	•	•	•	•	•	•	•	•
Total: 2 conditions $\times$ 4 rounds $\times$ 3 states $=$ 24 decisions per participant									

Notes: **P** = private game, **A** = audience game. Each • denotes one repayment decision over $\{0, 5{,}000, \ldots, 35{,}000\}$ Taka for a fixed debt obligation of 35,000 Taka. In the strategy method, participants choose a repayment amount for each income state within a game before any state is realized; the order in which the states are presented is randomized at the session level. Round 1 is the practice round, so 18 of the 24 decisions are incentivized. After all decisions are complete, one round–state pair is drawn at random within each game condition from the incentivized rounds; residual income from the two selected decisions is converted at 1/100 and added to a 150 Taka participation payment.

After all repayment decisions are completed in the session, one round–state pair is randomly drawn from each game condition to determine post-game earnings. The income state is determined by drawing a colored ball, with the proportions of balls representing the aforementioned probabilities for the states of the world. Income remaining after repayment in the selected private and audience decisions is converted at a rate of 1/100. Participants also receive a 150 Taka participation payment, ensuring a positive payment regardless of the selected repayment decisions. The conversion rate was chosen so that expected experimental earnings were approximately equivalent to a day's agricultural labor wage.

The design abstracts from dynamic incentives in the credit contract, that is, repayment in one round does not affect borrowing in any subsequent rounds. Game facilitators also ensured that no loan officers were present during the sessions. Thus, differences in repayment across game types should not be attributed to concerns about contractual pecuniary incentives or the presence of authoritative figures. Instead, in the audience games, borrowers expects their peers and neighbors to learn whether and how much they defaulted. By varying social observability within subjects while holding constant loan size, income risk, and monetary incentives, the experiment isolates the audience effect on repayment.

3.3 Sampling and Data Collection

The experiment was conducted between January and April 2025 with active BRAC microfinance borrowers in southern Bangladesh. Data collection was organized and implemented by the Brac Institute of Governance and Development (BIGD) in collaboration with BRAC-NGO and the University of California, Davis. Participants were drawn from a broader randomized control trial evaluating indexed risk-management tools among agricultural microfinance borrowers. The broader RCT includes 6,683 borrowers from 255 BRAC branches, local offices that administer microfinance lending, across 13 districts (Figure A.1). Branches were randomly selected within districts, and village organizations (VOs), groups of borrowers from the same local community who meet for loan repayments and savings, were randomly selected within agriculturally intensive areas. Within selected VOs, a screening questionnaire was administered to select farm households. BRAC offers small individual-liability microfinance loans only to female members, so both the broader RCT population and the experimental sample drawn from it consist entirely of women. Table A.1 reports sample characteristics, including borrower demographics, farming activities, credit access, recent shock exposure, and experience with crop insurance. The characteristics are drawn from the baseline survey of the broader RCT.

For the repayment experiment, I randomly selected 120 branches from the broader RCT, stratified by its three treatment arms, and randomly selected one VO within each branch. All agricultural borrowers in the broader RCT sample belonging to the selected VO were invited to participate in the experimental session. Approximately 12 borrowers were invited to the sessions included in the final experimental design, of whom 919 attended and participated, corresponding to an attendance rate of 78.4 percent.

Each session included 6 to 10 borrowers, with an average of approximately nine participants, drawn from an existing VO. Participation was voluntary, and no additional sample restrictions were imposed based on borrower characteristics or repayment behavior. The experimental sample therefore consists of agricultural microfinance borrowers from BRAC.

The study initially targeted 120 VOs and the first 20 sessions used an earlier version of the design with a single round, and a fixed ordering of the Private and Audience games. I exclude these sessions because as they do not share the repeated round and randomized game sequence structure of the final design. The analysis sample consists of 919 borrowers from the remaining 100 VOs, all of whom participated in the four-round design described in Section 3.2. Prior to field implementation, 13 pilot and facilitator-training sessions were conducted with members of the research team present to observe implementation and provide structured feedback. Across the pilot, initial, and final designs, 133 sessions were conducted at 120 village locations. Figure A.2 summarizes the resulting experimental sample.

Sessions were administered by nine trained BIGD field supervisors with prior experience conducting household surveys and coordinating enumerator teams in the study region. All held at least a bachelor's degree and received training in the experimental procedures before field implementation. The study protocol was approved by the Institutional Review Board of the BRAC James P Grant School of Public Health, BRAC University (Protocol IRB-2024-ES-37), and all participants provided oral informed consent before the experiment.

4 Empirical Strategy

I estimate the audience effect on repayment using within-subject variation between private and audience game conditions, $g \in \{P, A\}$, across three income states, $k \in \{G, N, B\}$, representing good, normal, and bad states. Let y^k_{igr} denote repayment by borrower i in each of rounds $r \in 1, \ldots, 4$ for each game condition g and income state k. The associated state probabilities for each k are $p^G = 0.30$, $p^N = 0.40$, and $p^B = 0.30$. Expected repayment is defined as the probability-weighted repayment across states, $y^E_{igr} = \sum_{k \in \{G,N,B\}} p^k y^k_{igr}$. For simplicity, let y_{igr} denote repayment in a given income state. I exclude the first round, which was introduced as an unincentivized practice round to familiarize participants with the repayment task. The analysis uses the remaining three rounds $r \in \{2, 3, 4\}$ and contains 18 observations per borrower across two game conditions, three rounds, and three income states.

4.1 Fixed effects estimation of observed repayment

In the OLS specification with fixed effects, identification of the audience effect comes from within-borrower variation in repayment between the game conditions,

$$y_{igr} = \beta \, Audience_g + \mu_i + \lambda_r + \varepsilon_{igr}, \tag{6}$$

where $Audience_g$ equals one for the audience game and zero for the private game. Individual fixed effects μ_i absorb time-invariant heterogeneity across borrowers, including baseline repayment propensities, socioeconomic characteristics, and session-specific factors. Round fixed effects λ_r account for common changes over the experiment, such as learning or fatigue. Standard errors are clustered at the session level to allow for heteroskedasticity and correlation across repeated decisions by the same borrower and across borrowers sharing the experimental environment, facilitator, and game sequence assignment. The coefficient β measures the within-borrower difference in observed repayment between the two game conditions in each state ($y_{igr} - \bar{y}_i = \beta(Audience_g - \overline{Audience_i}) + (\lambda_r - \bar{\lambda}_i) + (\varepsilon_{igr} - \bar{\varepsilon}_i)$), which I refer to as the audience effect of making default observable to peers.

Although individual fixed effects (μ_i) are not separately identified, the borrower-specific heterogeneity they absorb can meaningfully reflect the intrinsic behavioral motivations. As discussed in the conceptual framework (Section 2), borrowers may differ in the repayment norms they hold and the value they place on adhering to these norms, depending on their socioeconomic backgrounds, past experiences and expectations. In the private game, borrowers face a tradeoff between the monetary gain from defaulting and their intrinsic motivation to repay, without any external penalty. The borrower-specific heterogeneity in private game repayment, thus, is informative about the intrinsic behavioral motivations under the assumptions of the behavioral model.

4.2 Two-limit Tobit estimation of latent repayment

The distribution of repayment choices in Figure A.6 shows mass at both boundaries, with approximately 12 percent of observations at zero and 35 percent at the upper limit of $\overline{y} = 35{,}000$ Taka. The censoring may attenuate the observed audience effect relative to the underlying latent repayment response. To account for this, I estimate a two-limit Tobit model that treats boundary choices as censored realizations of latent repayment and jointly models the probabilities of choosing either boundary and repayment in

the interior (Wooldridge, 2002). The latent repayment equation and corresponding censoring rule are

$$
\begin{aligned}
y^*_{igr} &= \beta^T \, Audience_g + \mathbf{x}'_{igr}\boldsymbol{\psi} + \varepsilon^*_{igr}, \\
y_{igr} &= \begin{cases} 0 & \text{if } y^*_{igr} \leq 0, \\ y^*_{igr} & \text{if } 0 < y^*_{igr} < \overline{y}, \\ \overline{y} & \text{if } y^*_{igr} \geq \overline{y}. \end{cases}
\end{aligned} \tag{7}
$$

where $\mathbf{x}_{igr}$ includes round and game sequence indicators. I estimate the model by maximum likelihood, assuming normally distributed errors with constant variance. Standard errors are clustered at the session level, as in the fixed effects specification. The coefficient β^T measures the audience effect on latent repayment.

Unlike the OLS specification, the Tobit model does not include individual fixed effects. Although omitted individual effects may bias estimates when correlated with the regressors (Mundlak, 1978), each borrower makes the same number of decisions in both conditions, and the audience indicator is uncorrelated with time-invariant borrower characteristics by design. The pooled Tobit estimate nevertheless relies on the distributional assumptions imposed on the composite latent error.

Monte Carlo assessment of estimators. Motivated by the conceptual framework, I assume that the audience effect reflects an underlying behavioral response that may not be fully observed when repayment is bounded by the experimental choice set. The constant two-limit Tobit specification accounts for the game specific censored outcomes. However, borrowers choosing the same boundary in both game conditions may have an unobserved latent audience response despite showing no difference in observed repayment. This motivates a variable-limit Tobit specification for the difference between audience and private repayment, with censoring limits determined by private-game repayment.

Appendix D compares the finite-sample performance of the fixed effects, two-limit Tobit, and variable-limit Tobit estimators using simulations calibrated to the experimental design, varying borrower heterogeneity and dependence across repeated decisions. The results show that the fixed effects estimator recovers the audience effect on observed repayment, while the two-limit Tobit recovers the latent effect under the maintained censored-normal data-generating process. I also assess a variable-limit Tobit specification in the Monte Carlo simulations but exclude it from the main analysis because when the censoring limits depend on observed private repayment, which is itself censored and leads to overestimation of the latent effect in the simulations. These findings support interpreting the fixed effects and constant two-limit Tobit estimates as distinct measures of observed and underlying behavioral responses to the audience condition.

5 Results

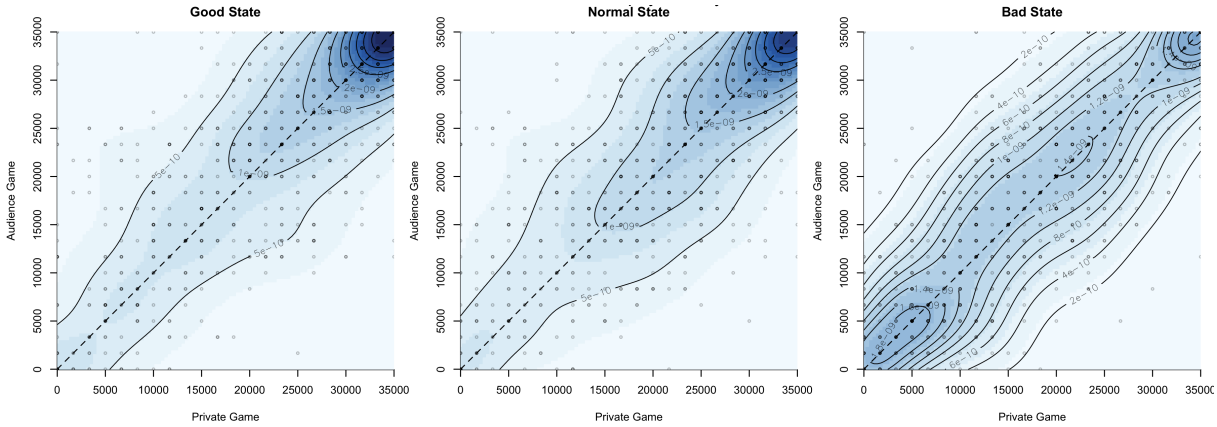
This section summarizes the main results. I present descriptive evidence on repayment across game conditions and income states, followed by fixed effects estimates of the audience effect on observed repayment, and two-limit Tobit estimates of the latent effect, the implied effect on observed repayment, and the probabilities of repayment at either bound. Finally, I assess the experimental design and examine whether and how repayment and the audience effect vary across income states.

5.1 Descriptive Evidence

The conceptual framework predicts higher repayment in favorable income states than in adverse states. Figure 5.1 shows the joint density of mean private and audience repayment, averaged across rounds, while Table A.2 reports average repayment by round, game condition, and income state. In the good and normal states, the density is concentrated near the upper-right corner, indicating that many borrowers repay close to the full debt obligation in both games. This is consistent with intrinsic repayment motivation even in the absence of an audience. In the bad state, where full repayment leaves borrowers with no residual income, this concentration dissipates, with two distinct modes near the lower and upper repayment bounds. While some borrowers repay very little, others repay nearly the full obligation even in the bad state of the world.

In the good and normal states, the ridge of the joint density lies slightly above the 45-degree line, which provides descriptive evidence of a positive audience effect. The distributions also reveal heterogeneity in repayment across borrowers in both games. As discussed in the conceptual framework, this heterogeneity may reflect differences in borrowers' perceptions of repayment norms and the value they place on adhering to or defying them.

Figure 5.1: Joint distribution of mean private game and audience game repayments.



5.2 Fixed Effects Estimates

Table 5.1 reports fixed effects estimates of the effect of social observability on repayment separately for each realized income states. Audience increases observed repayment by 562 to 798 Taka across income states. In the conceptual framework, these estimates provide reduced form evidence of the average audience effect, $\mathbb{E}[\Delta_i]$ relecting reputational motivation. The estimated effect is smallest in the bad state, at 562 Taka, and larger in the good and normal yield states, at 767 and 798 Taka respectively. Relative to repayment without an audience, these effects correspond to approximately 3.3 to 3.5 percent. The audience effect is positive and statistically significant in all states. Since monetary incentives are held fixed across the private and audience games, the positive audience effect reflects participants giving up additional earnings in response to the social observability of default. Appendix Table A.3 reports estimates restricted to rounds 2 and 3 and specifications using individual-level clustering, both of which are consistent with the main results.

Repayment in the private game varies substantially across states of nature. Average repayment is lowest in the bad yield state and higher in the normal and good yield states. In the conceptual framework, these estimates reflect changes in borrowers' intrinsic motivation to repay as the income state changes. A lower income realization increases the marginal utility of retained income, so repayment becomes more costly at the margin. It may also lower the borrower's intrinsic belief about what she feels obligated to repay given competing needs under adverse circumstances. Whether private repayment and the audience effect vary across income states is formally tested in Section 5.4.

Table 5.1: Audience Effect on Repayment: Fixed effects Estimates

	(1)	(2)	(3)	(4)
	Expected Repayment	*Good Yield Repayment*	*Normal Yield Repayment*	*Bad Yield Repayment*
Audience	717.99***	767.14***	797.97***	562.21***
	(147.81)	(197.70)	(248.96)	(198.45)
Constant	21,162.86***	23,108.45***	22,629.67***	17,261.52***
	(73.91)	(98.85)	(124.48)	(99.22)
Effect (% of Private Mean)	3.39%	3.32%	3.53%	3.26%
Observations	5,514	5,514	5,514	5,514
R-squared	0.80	0.69	0.66	0.73
Fixed Effects	Individual, Round	Individual, Round	Individual, Round	Individual, Round
SE Cluster	Session	Session	Session	Session

Notes: Standard errors clustered at the session level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3 Two-limit Tobit Estimates

Table 5.2 reports two-limit Tobit estimates that account for censoring at zero and the contractual repayment obligation of 35,000 Taka. Panel A reports the latent audience effect on repayment. The latent audience effect is positive and statistically significant across all outcomes, ranging from 794 Taka in the

bad yield state to 1,605 Taka in the good yield state. Relative to latent repayment without an audience, these estimates correspond to increases of approximately 3.8 to 5.3 percent. Since observed repayment is bounded, particularly with substantial mass at the full repayment limit, the latent audience effect is larger than the observed audience effect in 5.1.

Panel B reports the corresponding average marginal effects. Audience increases observed repayment by 768 Taka in the good state, 678 Taka in the normal state, and 494 Taka in the bad state, with an estimated increase of 676 Taka in expected repayment. Audience also increases the probability of full repayment by 1.1 to 2.6 percentage points across income states and reduces the probability of zero repayment by approximately 1.0 to 1.3 percentage points. For expected repayment, the corresponding changes are a 1.65 percentage-point increase in the probability of full repayment and a 0.62 percentage-point decline in the probability of zero repayment. Thus, the Tobit estimates indicate that audience shifts repayment both within the repayment bounds and toward full repayment.

Table 5.2: Audience Effect on Repayment: Two-Limit Tobit Estimates

	(1) *Expected Repayment*	(2) *Good Yield Repayment*	(3) *Normal Yield Repayment*	(4) *Bad Yield Repayment*
Panel A: Coefficients				
Audience	862.59***	1,605.15***	1,247.44***	793.76**
	(183.06)	(413.00)	(465.72)	(309.15)
Constant	22,756.09***	30,600.45***	27,374.22***	18,834.19***
	(916.31)	(1,650.14)	(1,514.20)	(1,272.94)
Panel B: Average Marginal Effects				
$\frac{\partial}{\partial x} E[y^* \mid x]$ (latent, unconstrained)	862.6***	1,605***	1,247***	793.8**
	(183.1)	(413.0)	(465.7)	(309.2)
$\frac{\partial}{\partial x} E[y \mid x]$ (observed, censored)	675.5***	768.2***	678.3***	494.1***
	(142.8)	(193.5)	(251.1)	(190.6)
$\frac{\partial}{\partial x} \Pr[y = \overline{y} \mid x]$ (p.p.)	1.653***	2.569***	2.187***	1.112***
	(0.354)	(0.643)	(0.810)	(0.431)
$\frac{\partial}{\partial x} \Pr[y = 0 \mid x]$ (p.p.)	−0.623***	−1.269***	−1.037***	−1.052**
	(0.153)	(0.337)	(0.394)	(0.413)
Latent Effect (% of Private Mean)	3.79%	5.25%	4.56%	4.21%
Observations	5,514	5,514	5,514	5,514
Uncensored	4,331	2,608	2,958	3,383
Left-censored	281	626	581	932
Right-censored	902	2,280	1,975	1,199
Controls	Round, Sequence	Round, Sequence	Round, Sequence	Round, Sequence
SE Cluster	Session	Session	Session	Session

Notes: Standard errors are clustered at the session level. Censoring limits are $[0, 35{,}000]$ Taka. All specifications include round and game sequence controls. Panel B probability effects are reported in percentage points. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.4 Is Repayment State Contingent?

The state-specific estimates in Tables 5.1 and 5.2 suggest lower private repayment in adverse income states and a somewhat smaller audience effect in the bad state. To test the differences across the states, I stack repayment decisions across the three income states and interact the audience indicator with state indicators,

$$y_{igr}^{k} = \beta \, Audience_g + \sum_{j \in \{N,B\}} \delta^j \left(\mathbf{1}\{k = j\} \times Audience_g \right) + \sum_{j \in \{N,B\}} \psi^j \mathbf{1}\{k = j\} + \mu_i + \lambda_r + \varepsilon_{igr}^{k}. \quad (8)$$

where the good state in the private game is the reference category. The coefficients ψ^j measure differences in private repayment relative to the good state, while β captures the audience effect in the good state. The interaction coefficients δ^j measure whether the audience effect differs in the normal and bad states relative to the good state.

Table 5.3 reports the estimates of state dependence in repayment and the audience effect. Relative to the good state, private repayment is 479 Taka lower in the normal state and 5,847 Taka lower in the bad state. The former difference is marginally significant, while the latter is large and statistically significant. This state dependence is consistent with the conceptual framework, in which the marginal utility of retained income varies across income states, and intrinsic repayment norms may adjust to borrowers' economic circumstances and competing needs.

Interestingly, the audience effect does not vary detectably across income states, despite substantial state dependence in private repayment. The estimated audience effect in the good state is 767 Taka, with differences of 31 Taka in the normal state and -205 Taka in the bad state, neither of which is statistically significant. The two-limit Tobit estimates yield similar results for latent repayment. Relative to the estimated latent audience effect of 1,497 Taka in the good state, the interaction coefficients are -239 Taka in the normal state and -676 Taka in the bad state, neither statistically significant.

This finding is particularly important given the information structure of the experiment. Borrowers were informed that peers would observe their default amount but *not* the income state in which the repayment decision was made. Thus, borrowers could not rely on the disclosed information to justify lower repayment under adverse income conditions. The absence of detectable state dependence in the audience effect is consistent with a reputational response driven by social observability of default and adds repayment incentives to the state-dependent intrinsic motivation rather than substituting for it.

Table 5.3: State Dependence of Repayment and the Audience Effect

	(1) OLS FE	(2) Two-limit Tobit
Normal	−478.78*	−1,267.60**
	(286.84)	(564.83)
Bad	−5,846.94***	−10,059.95***
	(481.98)	(821.83)
Audience	767.14***	1,496.52***
	(197.69)	(379.93)
Audience × Normal	30.83	−239.20
	(293.80)	(549.39)
Audience × Bad	−204.93	−676.49
	(257.34)	(449.96)
Constant	23,108.45***	29,149.24***
	(232.77)	(1,408.39)
Observations	16,542	16,542
R-squared	0.592	0.003 (Pseudo)
Fixed Effects / Controls	Individual, Round	Game sequence, Round
Left-censored	—	2,139
Right-censored	—	5,454
Censoring Limits	—	[0, 35,000]

Notes: The omitted category is the good state in the private game. The estimated latent audience effect in the good yield state is 1,497 Taka, compared with 1,605 Taka in the state-specific Tobit estimate reported in Table 5.2. The difference is due to the stacked specification imposing a common latent error variance across income states, whereas the state specific specifications allow this variance to differ by state. Standard errors are clustered at the session level. *** p < 0.01, ** p < 0.05, * p < 0.1.

5.5 Experimental Design Diagnostics

The within-subject design uses repeated repayment choices across rounds and experimental conditions. I first assess whether borrowers' relative repayment behavior is consistent across repeated decisions. I then examine whether the audience effect varies across rounds or depends on game order. Appendix C.1 shows substantial consistency in repayment choices within both the private and audience games. Standardized Cronbach's alpha ranges from 0.791 to 0.809 across income states and from 0.810 to 0.869 across rounds, with all coefficients well above those generated under random choice. Appendix C.2 then examines changes in repayment over rounds, variation in the audience effect across rounds, and differences by game order.

Temporal stability. A concern in a within-subject design is that repayment may change with position in the session. Appendix C.2 shows that repayment declines from round 2 to round 4 in every income state and under both experimental conditions, with round effects jointly different from zero in all three states. List (2025) notes that common time-varying factors can be accounted for using time fixed effects when they do not systematically alter the treatment effect. The main specifications therefore include round fixed effects. Appendix C.2 also shows that the round-by-audience interactions are jointly indistinguishable from zero in all three states, although they are imprecisely estimated. All decisions are made within a single session, and round order cannot be randomized. The round effects therefore

cannot be separated from fatigue, learning, or other changes over the session.

Causal transience. A further concern is that behavior under the current experimental condition may be affected by exposure to the other condition through anticipation or carryover (List, 2025). The design limits one important source of carryover by delaying disclosure until all repayment decisions are complete. Borrowers therefore do not observe peer reactions to disclosed default before making subsequent choices. This removes the peer-feedback channel through which an audience decision could directly affect later choices. The order of the private and audience games is randomized at the session level, which with two conditions corresponds to the two-period Latin square discussed by List (2025) and provides a diagnostic for order effects. Appendix C.2 shows no detectable difference in the audience effect across orders, although the estimates are not sufficiently precise to rule out moderate order effects. Anticipation remains a concern that cannot be tested with these data.

6 Heterogeneity in Repayment and the Audience Effect

The conceptual framework discusses repayment behavior and the response to social observability to vary across borrowers. This section examines whether private repayment and the audience effect vary with educational attainment, recent shock exposure, and the geographic dispersion of households within the borrowing group. The estimates characterize reduced-form heterogeneity in repayment behavior and do not separately identify the behavioral parameters underlying these differences.

6.1 Educational Attainment

To assess whether this effect varies with education, I replace individual fixed effects round fixed effects, which allows time invariant characteristics to enter as regressors alongside their interactions with the audience indicator. Because education is absorbed by individual fixed effects in the within-subject design, the estimates in Table 6.1 rely on between-borrower variation.

The estimated audience effect among borrowers with less than Grade 9 education is positive across all income states. Borrowers with Grade 9 or higher education report substantially lower repayment in private games, whereas interaction of audience with more schooling is larger and significant in the good, normal, and expected yield states, with estimates ranging from approximately 972 to 1,341 Taka, and is imprecisely estimated in the bad yield state.

Table 6.1: Heterogeneity in the Audience Effect by Educational Attainment

	(1) *Expected Repayment*	(2) *Good Yield Repayment*	(3) *Normal Yield Repayment*	(4) *Bad Yield Repayment*
Audience × Grade 9+	971.67**	1,341.35**	1,083.99*	452.22
	(428.23)	(594.77)	(635.68)	(541.45)
Audience	566.80***	558.42***	629.30**	491.84**
	(151.44)	(208.26)	(270.75)	(221.94)
Constant	21,162.86***	23,108.45***	22,629.67***	17,261.52***
	(73.19)	(97.79)	(123.84)	(99.22)
Observations	5,514	5,514	5,514	5,514
R-squared	0.798	0.693	0.660	0.727
Fixed Effects	Individual, Round	Individual, Round	Individual, Round	Individual, Round
SE Cluster	Session	Session	Session	Session

Standard errors clustered at the session level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.2 Flood Exposure and Household Dispersion

The conceptual framework allows a recent covariate shock to raise or lower the value of social standing, γ_i. Flood exposure may also be informally known within the VO because flooding is spatially correlated across nearby households. If peers can infer which borrowers were affected, explicit audience may add less information about the reason for lower repayment. I measure household dispersion as standardized distance of households in a VO. Table 6.2 examines how the audience effect varies with flood exposure and household dispersion.

Columns (1)– (3) show that flood exposure does not increase the audience effect in any state. The differential is statistically indistinguishable from zero in the normal and bad states, while in the good state flood exposure offsets most of the audience effect, reducing it from 1,112 to approximately 317 Taka. This pattern is consistent with flood exposure providing a justification for lower repayment when repayment would otherwise be relatively high. In the normal and bad states, the income realization may already provide such a justification, leaving less scope for an audience effect of flood exposure.

Columns (4)– (6) interact flood exposure with household dispersion. Among borrowers without recent flood exposure, dispersion is unrelated to the audience effect in the good and normal states and weakly negatively associated with it in the bad state. The triple interaction is positive in all three states and statistically significant in the normal and bad states, indicating that the difference in the audience effect between flood exposed and unexposed borrowers becomes more positive as household dispersion increases. Information about flood exposure being less readily inferred when households are more geographically dispersed, increasing the role of audience. The triple interaction is small and imprecisely estimated in the good state, where the offset associated with flood exposure does not vary detectably with household dispersion. These estimates are exploratory and identify the combined reduced-form

response rather than separate effects operating through γ_i or $n^R(\mathbf{x}_i)$.

Table 6.2: Heterogeneity in the Audience Effect by Flood Exposure and Household Dispersion

	(1) *Expected Repayment*	(2) *Good Yield Repayment*	(3) *Normal Yield Repayment*	(4) *Bad Yield Repayment*
Audience × Shock × HH dispersion	700.58**	238.25	1,030.79**	722.64**
	(309.12)	(491.94)	(484.30)	(347.30)
Audience × HH dispersion	−117.40	−103.55	−47.30	−224.72*
	(129.84)	(135.57)	(291.58)	(135.33)
Audience × Shock	−211.81	−789.59**	397.65	−446.65
	(289.64)	(392.69)	(444.00)	(371.02)
Audience	838.16***	1,119.55***	666.83**	785.22***
	(206.84)	(277.17)	(330.44)	(293.09)
Constant	21,162.86***	23,108.45***	22,629.67***	17,261.52***
	(72.99)	(97.14)	(122.87)	(98.05)
Observations	5,514	5,514	5,514	5,514
R-squared	0.798	0.692	0.660	0.727
Fixed Effects	Individual, Round	Individual, Round	Individual, Round	Individual, Round
SE Cluster	Session	Session	Session	Session

Standard errors clustered at the session level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.3 Perceived Repayment Norms

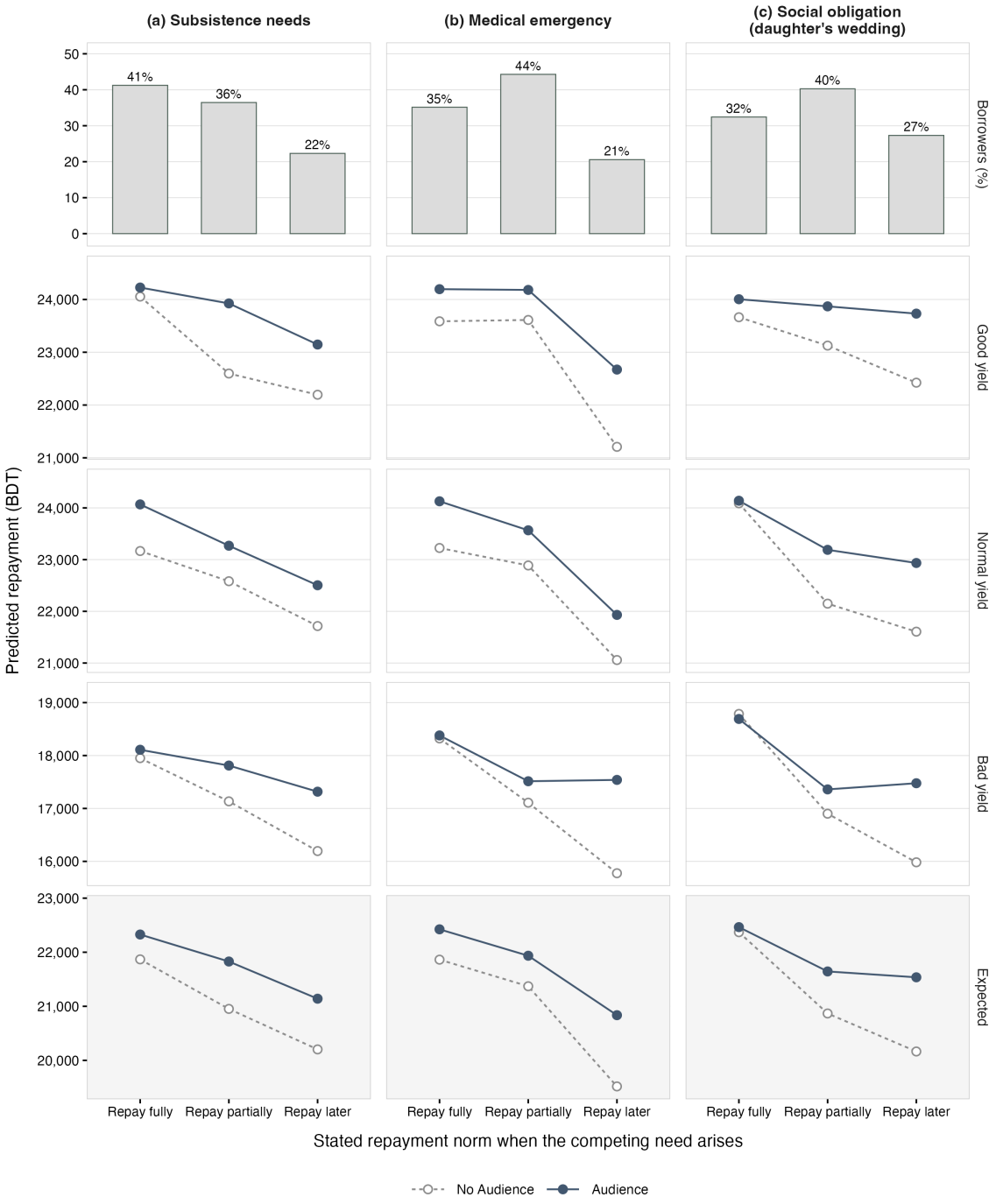
We ask borrowers what they believe they should repay when faced with competing needs such as subsistence expenses, a medical emergency, and a social obligation (dowry for daughter's wedding). For each situation, borrowers indicated whether they should repay on time fully, partially, or delay until they have the opportunity to repay. The stated norms were elicited from borrowers accustomed to microfinance practices in which default is generally socially observable. Their responses may therefore reflect both their own beliefs about appropriate repayment, $n^I(\theta_i)$, and what they believe others expect of them, $n^R(\mathbf{x}_i)$. I hypothesize that borrowers who consider lower or delayed repayment appropriate would repay less in both private and audience games, but may nevertheless respond positively to an audience. The response may depend on whether borrower believes the competing household needs could be justified to the peers.

Figure 6.1 shows that these beliefs are associated with repayment behavior. Across all three scenarios, borrowers who consider partial or delayed repayment appropriate repay less in both private and audience games. The differences are observed across income states, suggesting that repayment decisions are associated with what borrowers consider appropriate repayment norms, even when the experimental income and debt obligation are held fixed.

Whether these beliefs are also associated with the audience effect is less uniform. Table A.5 shows some evidence of heterogeneity for subsistence needs in the good state and medical emergencies in the bad state. In the latter case, borrowers who consider delaying repayment appropriate have an audience

effect 1,707 Taka more ($p = 0.01$) than those who believe they should repay fully. The differences are more pronounced for social obligations. Borrowers who consider delaying repayment for a daughter's wedding appropriate have an audience effect approximately 1,278 Taka more in the normal state ($p = 0.04$) and 1,589 Taka more in the bad state ($p < 0.01$) than those who believe they should repay fully. The interactions are jointly significant in the normal and bad states and for expected repayment.

Figure 6.1: Heterogeneity in the Audience Effect by Perceived Repayment Norms



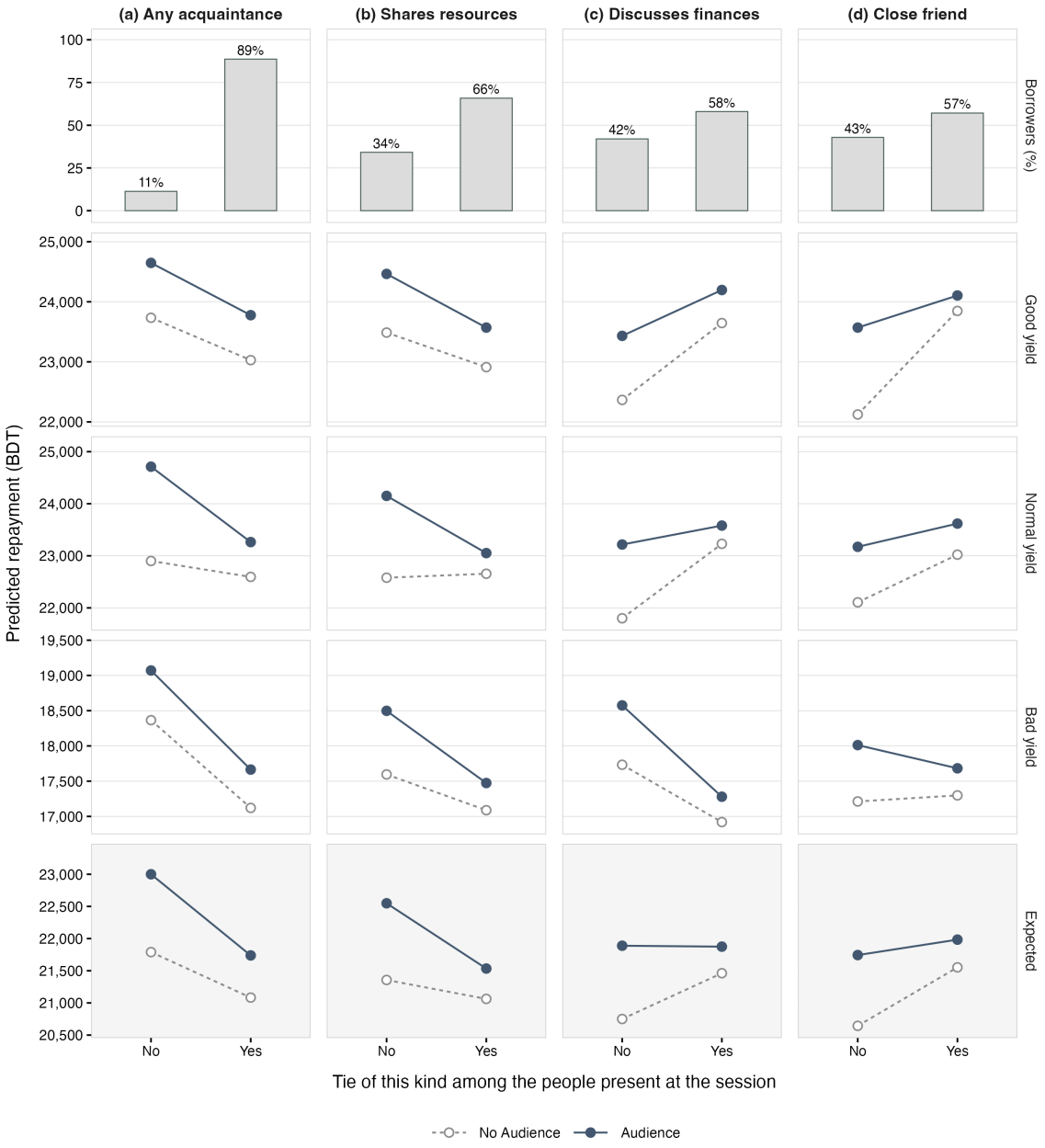
6.4 Social Ties with the Audience

The value individuals place on adhering to intrinsic and reputational norms, α_i and γ_i, may depend on their social relationships (Breza and Chandrasekhar, 2019; Bursztyn and Jensen, 2015). During sessions at village centers, neighbors were not restricted from observing the proceedings along with the members of the borrowing groups. This allowed us to examine heterogeneity by the type of social ties with the audience. We asked borrowers whether anyone present was an acquaintance, someone with whom they shared resources or discussed finances, or a close friend. Borrowers with closer social ties may expect their peers to be more understanding of default, potentially reducing the additional reputational pressure created by explicit observability.

Figure 6.2 shows that 89% of borrowers had at least one acquaintance, 66% someone with whom they shared resources, 58% someone with whom they discussed finances, and 57% a close friend among the audience. Borrowers with acquaintances or resource-sharing ties in the audience repay less in both private and audience games across all income states. Whereas, those who discuss finances or have close friends in the audience generally repay more in the good and normal states, but this pattern does not hold in the bad state.

Table A.6 shows that the audience effect is generally smaller when borrowers have these social ties with someone in the audience. For expected repayment, the audience effect is 720 Taka smaller ($p = 0.03$) among borrowers who share resources with someone in the audience, 726 Taka smaller ($p = 0.02$) among those who discuss finances, and 669 Taka smaller ($p = 0.04$) among those with a close friend. By income state, the interactions are significant in the normal state for resource-sharing and financial-discussion ties and in the good state for close friendships. None is significant in the bad state. For acquaintances, the interaction is negative but not statistically significant in any income state or for expected repayment.

Figure 6.2: Heterogeneity in the Audience Effect by Social Ties



7 Interpreting the Magnitude in Lender Profitability

Microfinance institutions face substantial transaction and operating costs associated with administering and collecting many small loans (Cull et al., 2009; Rosenberg et al., 2013). Even a small increase in realized repayment can therefore have a substantial effect on lender net earnings when the lending margin is small relative to the amount intermediated. This section illustrates this relationship using a simple accounting calculation calibrated to BRAC's audited microfinance accounts.

For each taka of principal, let i_L denote the lending rate, i_B the capital cost, and c the nonfinancial operating cost. The contractual amount due is $\bar{r} = 1 + i_L$. If $d \in [0, 1]$ denotes the default share, defined

as the unpaid share of the contractual amount, the lender collects $(1-d)\bar{r}$. Net earnings per unit of principal are

$$\pi(d) = \underbrace{(1-d)(1+i_L)}_{\text{collections}} - \underbrace{(1+i_B)}_{\text{capital cost}} - \underbrace{c}_{\text{operating cost}}. \tag{9}$$

Capital and operating costs do not vary with repayment in this accounting exercise, so the lender bears the cost of the default share. Under full repayment, net earnings are $\pi(0) = i_L - i_B - c$. Differentiating equation (9) gives $\partial\pi/\partial d = -(1+i_L)$. A one percentage point increase in the default share therefore reduces net earnings by $(1+i_L)/100$ per unit of principal. Relative to earnings under full repayment, the amplification factor is,

$$\frac{-\partial\pi/\partial d}{\pi(0)} = \frac{1+i_L}{i_L - i_B - c}. \tag{10}$$

where the break even default share is $d^* = \frac{i_L - i_B - c}{1+i_L}$.

I calibrate these parameters using BRAC's microfinance accounts audit for fiscal year 2024–25 (BRAC, 2025) and report the calculation in terms of both principal disbursed and average principal outstanding.[3] In terms of principal disbursed, the calibration uses $i_L = 0.130$, $i_B = 0.033$, and $c = 0.057$. This corresponds most closely to the one-cycle structure of equation (9). It gives $\pi(0) = 0.039$ and a break even default share of approximately 3.5 percent. For comparison, in terms of average principal outstanding, the corresponding values are $i_L = 0.227$, $i_B = 0.058$, and $c = 0.100$, with $\pi(0) = 0.069$ and a break even default share of approximately 5.6 percent. The latter is directly comparable to portfolio yields and expense ratios commonly reported for microfinance institutions. Table 7.1 reports net earnings under alternative default shares using both calibrations. In the experiment, an increase in

Table 7.1: Net Earnings under Alternative Default Shares

	Principal Disbursed		Average Principal Outstanding	
Default share d	Net earnings $\pi(d)$	Loss (% of $\pi(0)$)	Net earnings $\pi(d)$	Loss (% of $\pi(0)$)
0.0%	0.0394	0	0.0688	0
0.5%	0.0338	14	0.0626	9
1.0%	0.0281	29	0.0565	18
1.93%[a]	0.0176	55	0.0451	34
2.0%	0.0168	57	0.0442	36
3.0%	0.0055	86	0.0320	54
3.5%	0.0000	100	0.0258	62
5.6%	−0.0239	161	0.0000	100

Notes. Calculated from equation (9) using parameters calibrated to BRAC's audited microfinance accounts for fiscal year 2024–25 (BRAC, 2025). The first calibration is expressed in terms of principal disbursed; the second is expressed in terms of average principal outstanding. [a]The 1.93 percent row corresponds to the magnitude of the experimental-equivalent reduction in the default share discussed in the text. Because net earnings are linear in d, the earnings loss shown at this default share is also the earnings gain associated with a 1.93 percentage-point reduction in the default share.

[3]BRAC reports microfinance service-charge income of 93.7 billion Taka, financial expenditure of approximately 24.0 billion Taka, and total operating expenditure of approximately 65.3 billion Taka in fiscal year 2024–25. Nonfinancial operating expenditure is therefore approximately 41.3 billion Taka. In terms of total principal disbursed of 721.1 billion Taka, these give $i_L = 0.130$, $i_B = 0.033$, and $c = 0.057$. Average principal outstanding is approximately 413.5 billion Taka, calculated from 397.0 billion Taka at the beginning of the year and 429.9 billion Taka at the end of the year. In terms of average principal outstanding, $i_L = 0.227$, $i_B = 0.058$, and $c = 0.100$. Loss provisions are not included in c because repayment losses are represented directly by the default share d in equation (9).

observed repayment of Δr corresponds to a reduction in the default share of the contractual obligation of $\Delta d = -\Delta r/\bar{r}$. For example, the estimated increase in observed expected repayment of 675.5 Taka against the debt obligation of 35,000 Taka corresponds to a 1.93 percentage-point reduction in the default share of the experimental obligation. If an equivalent reduction in the default share occurred on an otherwise comparable field loan, the BRAC calibrations imply an increase in lender net earnings equivalent to approximately 55 percent of full-repayment earnings in terms of principal disbursed and 34 percent in terms of average principal outstanding.

8 Conclusion

Microfinance institutions retain group-based repayment practices after moving from joint to individual liability that make delinquency observable to peers. This paper examines whether this remaining socially embedded feature affects repayment. Using variation in whether default is observable to peers, that is, whether the repayment decision is made before an audience, I find that social observability raises observed repayment by 3.3 to 3.5 percent and latent repayment by 3.8 to 5.3 percent. Since the experiment holds monetary incentives fixed across conditions and removes consequences for future credit access, these estimates isolate the audience effect on repayment from other contractual enforcement mechanisms. The shift to individual liability therefore removes shared financial responsibility without necessarily eliminating socially embedded repayment incentives.

Repayment without an audience is state dependent as predicted by model. Borrowers repay between one-half and two-thirds of the debt obligation, with repayment falling in the adverse income state even though full repayment remains feasible. In contrast, the audience effect does not vary detectably across income states. This finding is consistent with the interpretation that private repayment adjusts to realized economic circumstances, while social observability induces an additional repayment reflecting reputational motivation across income states.

An important feature of the experiment is that peers would observe the borrower's default amount but not the income state in which that decision was made. The absence of detectable state dependence in the average audience effect is consistent with this information structure. The exploratory heterogeneity results nevertheless suggest that peers may draw on information beyond what the lender explicitly reveals. Recent flood exposure offsets much of the audience effect in the good state, consistent with prior hardship providing a plausible justification for lower repayment even when current income is favorable. In the normal and bad states, the interaction between flood exposure and the audience effect becomes more positive as households within the borrowing group are more geographically dispersed. One possible interpretation is that audience matters more when peers have less direct information about one another's circumstances. These estimates do not separately identify this informational mechanism,

but they suggest that the effect of observable default may depend partly on what peers can otherwise infer about the circumstances surrounding repayment.

The repayment gain is small relative to the total debt obligation but can matter for lenders when lending margins are thin. The estimated increase in observed expected repayment of 675.5 Taka corresponds to a 1.93 percentage-point reduction in the default share. If a comparable reduction in field default occurred on an otherwise similar loan, the BRAC accounting calibration implies a profit gain equivalent to approximately 34–55 percent of the zero-default profit margin. This calculation illustrates why relatively small changes in repayment can have meaningful implications for lender profitability.

This profitability calculation does not, however, establish that strengthening social observability is welfare improving. It reflects a potential gain to the lender but does not account for nonpecuniary costs to borrowers. If reputational motivation makes borrowing more costly, socially observable repayment practices could affect credit demand even where microfinance relaxes conventional collateral requirements (Stiglitz and Weiss, 1981; Boucher et al., 2008). This is relevant in settings where households rely on the same social relationships for informal support and risk sharing.

The repeated repayment decisions provide a basis for studying borrower-level heterogeneity in repayment behavior and responses to social observability beyond the experiment. Choices are highly consistent within borrowers across states and rounds, suggesting persistent borrower-specific variation in repayment behavior. Future work can examine whether borrower-level measures constructed from private and socially observable repayment decisions predict borrowing, credit demand, and demand for risk-management products in the field (Castillo et al., 2018). A related extension would vary not only whether default is observable, but also what peers know about a borrower's repayment capacity, past behavior, and exposure to shocks. This could help explain whether the same default is judged differently depending on what peers know about the borrower, and how these perceptions affect the incentive to repay.

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A Appendix: Figures and Tables

Figure A.1: Study Site

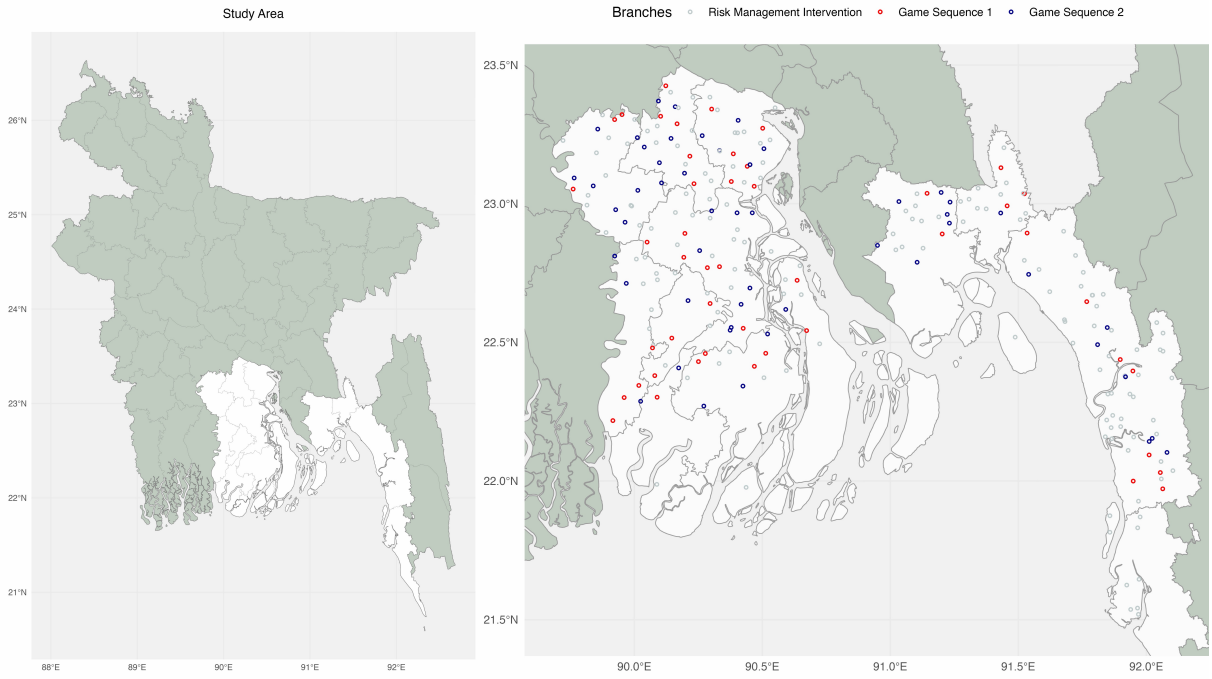


Figure A.2: Randomization of Game Design

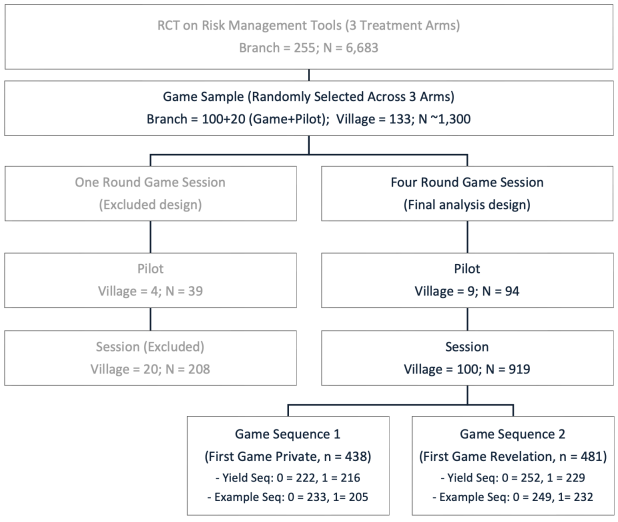


Table A.1: Experimental Sample Characteristics

	% or Mean	S.D.	*N*
Respondent characteristics			
Age in years	40.38	10.07	919
Grade 9 or higher education (%)	15.56		919
Food insecurity score	9.16	6.21	919
Risk preference	5.83	4.39	919
Farming type			
Rice farmer (%)	71.93		919
Non-rice farmer (%)	50.27		919
Credit			
Total BRAC loan (Taka)	67,310	60,434	919
Total loan (Taka)	199,098	208,558	919
Shock exposure			
Flood, last season (%)	43.42		919
Pest, last season (%)	17.63		919
Insurance & future interest in risk management tools			
Purchased crop insurance (%)	13.33		285
Interested in buying crop insurance (%)	79.95		848
Borrowing group information			
Participants in the experiment	9.19	.97	100
Household dispersion within VO (km)	0.254	0.234	100

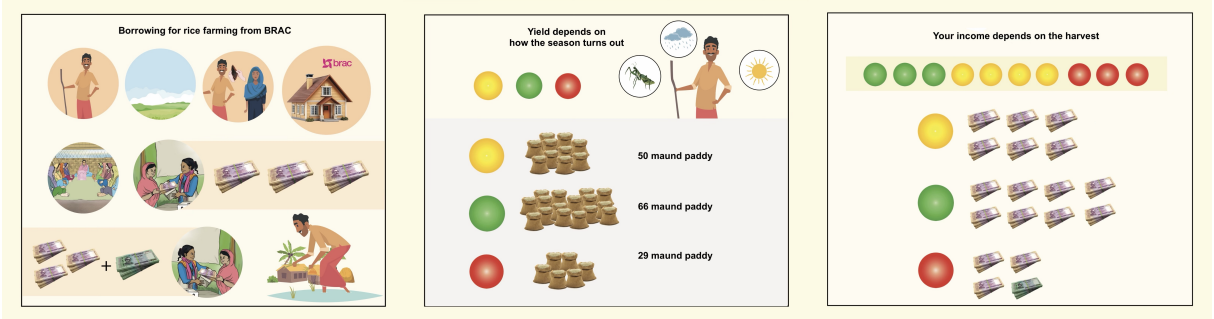
Notes: Binary variables are reported as percentages and continuous variables as means with standard deviations. Missing values in these descriptive characteristics do not affect the sample used in the main experimental repayment analysis.

Figure A.3: Actual village meeting and experimental session settings



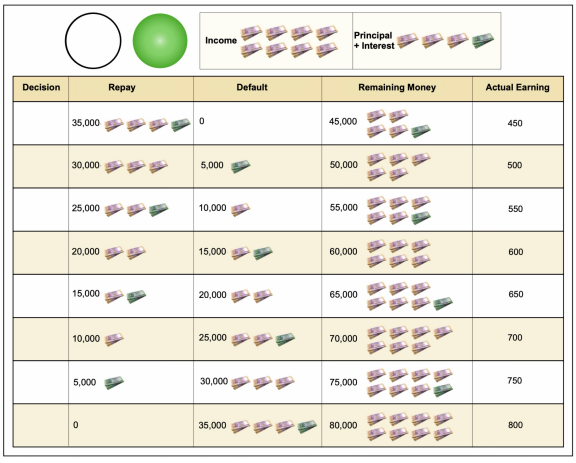
Notes: The left panel shows a BRAC village organization meeting, illustrating the group setting in which microfinance activities take place. The right panel shows an experimental session conducted at a village meeting location. The left-panel photograph is from BRAC.

Figure A.4: Visual aids for the experimental scenario



Notes: The visual aids were used to explain the rice-farming loan, uncertainty in harvest outcomes, and the corresponding income states. Illustrations of Bangladeshi Taka were used so that participants could follow the monetary amounts without relying on literacy.

Figure A.5: Decision sheet and board to reveal default for the audience game



Decision	Repay	Default	Remaining Money	Actual Earning
	35,000	0	45,000	450
	30,000	5,000	50,000	500
	25,000	10,000	55,000	550
	20,000	15,000	60,000	600
	15,000	20,000	65,000	650
	10,000	25,000	70,000	700
	5,000	30,000	75,000	750
	0	35,000	80,000	800

Name of the Borrower	Amount Defaulted
Farhana Akter	
Shyamoli Rani Das	
Mousumi Akter	
Nipa Begum	
Sharmin Sultana	
Monira Mondol	
Tania Begum	
Nahida Parvin	
Rukaiya Haque	
Sanjida Siddika	
Purnima Sarker	

Notes: The left panel shows the repayment decision sheet for the good income state. Similar decision sheets were used for corresponding normal and bad income states. Participants selected a repayment amount from the available choices, with the corresponding default, remaining income, and experimental earnings shown for each choice. Analogous decision sheets were used for the normal and bad income states, with amounts adjusted to the corresponding income levels. The right panel shows the board displayed in the audience game to indicate that the default amount from the randomly selected audience game would be revealed to members of the borrowing group.

Figure A.6: Repayment distribution across conditions and income states (round 2–4)

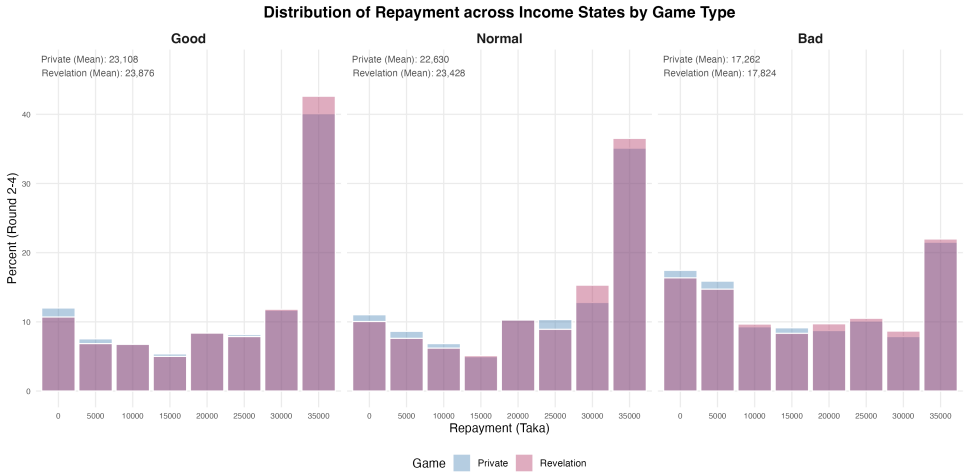


Table A.2: Descriptive statistics of repayment decisions, by round, audience, and income state

Round	Observability	Income state	*N*	Mean	S.D.	Min	p25	p50	p75	Max
1	Private	Good	919	27,149	11,584	0	20,000	35,000	35,000	35,000
		Normal	919	27,873	10,920	0	25,000	35,000	35,000	35,000
		Bad	919	21,616	12,945	0	10,000	25,000	35,000	35,000
	Audience	Good	919	27,998	10,965	0	25,000	35,000	35,000	35,000
		Normal	919	28,351	10,689	0	25,000	35,000	35,000	35,000
		Bad	919	22,378	12,848	0	10,000	25,000	35,000	35,000
2	Private	Good	919	23,781	12,699	0	15,000	30,000	35,000	35,000
		Normal	919	23,058	12,586	0	10,000	25,000	35,000	35,000
		Bad	919	18,112	13,181	0	5,000	20,000	30,000	35,000
	Audience	Good	919	24,374	12,488	0	15,000	30,000	35,000	35,000
		Normal	919	23,808	12,357	0	15,000	30,000	35,000	35,000
		Bad	919	18,939	12,784	0	5,000	20,000	30,000	35,000
3	Private	Good	919	22,894	13,067	0	10,000	30,000	35,000	35,000
		Normal	919	22,884	12,551	0	10,000	25,000	35,000	35,000
		Bad	919	17,035	12,899	0	5,000	15,000	30,000	35,000
	Audience	Good	919	23,993	12,598	0	15,000	30,000	35,000	35,000
		Normal	919	23,738	12,045	0	15,000	30,000	35,000	35,000
		Bad	919	17,845	13,046	0	5,000	20,000	30,000	35,000
4	Private	Good	919	22,650	13,035	0	10,000	30,000	35,000	35,000
		Normal	919	21,948	12,696	0	10,000	25,000	35,000	35,000
		Bad	919	16,638	12,824	0	5,000	15,000	30,000	35,000
	Audience	Good	919	23,259	12,957	0	10,000	30,000	35,000	35,000
		Normal	919	22,737	12,663	0	10,000	30,000	35,000	35,000
		Bad	919	16,687	12,784	0	5,000	15,000	30,000	35,000

Notes: Repayment amounts are in Bangladeshi Taka; the maximum feasible repayment is 35,000. Means and standard deviations are rounded to the nearest Taka.

Robustness to Clustering and Final-Round Exclusion

I examine two robustness checks for the main fixed effects estimates. First, the preferred specifications cluster standard errors at the session level to allow for correlation across decisions made within the same experimental session. Panel A in Table A.3 instead clusters standard errors at the borrower level. The standard errors and statistical significance are similar under the alternative clustering level, and the estimated audience effects are unchanged.

Second, I restrict the sample to rounds 2 and 3 to examine whether the main estimates are sensitive to behavior in the final experimental round. Panel B in Table A.3 shows that the audience effect remains positive and statistically significant across all income states and for expected repayment. The estimates are similar in magnitude to, and in some cases somewhat larger than, those obtained using rounds 2–4. The main results therefore do not appear to be driven by inclusion of the final round.

Table A.3: Robustness of the Audience Effect

	(1) *Expected Repayment*	(2) *Good Yield Repayment*	(3) *Normal Yield Repayment*	(4) *Bad Yield Repayment*
Panel A. Borrower-clustered standard errors				
Audience	717.99***	767.14***	797.97***	562.21***
	(145.82)	(211.20)	(221.73)	(198.31)
Constant	21,162.86***	23,108.45***	22,629.67***	17,261.52***
	(72.91)	(105.60)	(110.86)	(99.15)
Observations	5,514	5,514	5,514	5,514
Game Rounds	2–4	2–4	2–4	2–4
SE Cluster	Individual	Individual	Individual	Individual
Panel B. Excluding the final round				
Audience	820.46***	846.03***	802.50**	818.82***
	(167.99)	(246.57)	(307.10)	(259.55)
Constant	21,461.64***	23,337.87***	22,970.62***	17,573.45***
	(84.00)	(123.29)	(153.55)	(129.78)
Observations	3,676	3,676	3,676	3,676
Game Rounds	2–3	2–3	2–3	2–3
SE Cluster	Session	Session	Session	Session
Fixed Effects	Individual, Round	Individual, Round	Individual, Round	Individual, Round

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Audience Effect Heterogeneity

Table A.4: Heterogeneity in the Audience Effect by Flood Exposure

	(1) *Expected Repayment*	(2) *Good Yield Repayment*	(3) *Normal Yield Repayment*	(4) *Bad Yield Repayment*
Audience × Shock	−257.54	−794.72**	309.80	−476.83
	(291.39)	(385.10)	(448.08)	(375.05)
Audience	829.81***	1,112.18***	663.46**	769.23***
	(205.10)	(272.58)	(331.06)	(291.28)
Constant	21,162.86***	23,108.45***	22,629.67***	17,261.52***
	(73.85)	(97.42)	(124.02)	(98.56)
Observations	5,514	5,514	5,514	5,514
R-squared	0.798	0.692	0.659	0.727
Fixed Effects	Individual, Round	Individual, Round	Individual, Round	Individual, Round
SE Cluster	Session	Session	Session	Session

Standard errors clustered at the session level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.5: Heterogeneity in the Audience Effect by Perceived Repayment Norms

	(1) *Expected Repayment*	(2) *Good Yield Repayment*	(3) *Normal Yield Repayment*	(4) *Bad Yield Repayment*
Panel A. Subsistence				
Audience	459.54**	171.50	901.50**	158.31
	(197.00)	(292.43)	(350.25)	(294.69)
× Repay partially	416.58	1,156.85**	−214.93	518.31
	(292.29)	(454.85)	(487.14)	(418.26)
× Repay later	477.86	779.72	−112.88	963.64
	(395.78)	(534.36)	(672.87)	(599.41)
p (interactions = 0)	0.24	0.04	0.91	0.22
Panel B. Medical emergency				
Audience	560.89**	608.88*	902.99**	56.76
	(247.39)	(342.75)	(389.30)	(335.43)
× Repay partially	3.40	−39.67	−223.22	348.65
	(340.17)	(496.87)	(536.12)	(433.92)
× Repay later	756.57	854.97	−29.98	1,706.91**
	(527.80)	(715.49)	(733.22)	(665.63)
p (interactions = 0)	0.28	0.41	0.90	0.03
Panel C. Social obligation				
Audience	93.96	341.16	50.34	−95.08
	(227.83)	(329.03)	(352.08)	(321.01)
× Repay partially	683.07**	402.08	990.20**	554.54
	(279.43)	(453.31)	(451.59)	(437.72)
× Repay later	1,277.89***	966.94	1,277.69**	1,589.10***
	(396.09)	(617.48)	(615.20)	(523.92)
p (interactions = 0)	<0.01	0.29	0.05	0.01
Observations	5,514	5,514	5,514	5,514
Fixed Effects	Individual, Round	Individual, Round	Individual, Round	Individual, Round
SE Cluster	Session	Session	Session	Session

Standard errors clustered at the session level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.6: Heterogeneity in the Audience Effect by Social Network Ties

	(1) *Expected Repayment*	(2) *Good Yield Repayment*	(3) *Normal Yield Repayment*	(4) *Bad Yield Repayment*
Panel A. Any acquaintance				
Audience	1,209.94**	913.46	1,810.90***	705.13
	(484.40)	(657.70)	(673.59)	(508.19)
× Any acquaintance	−554.72	−165.00	−1,142.19	−161.16
	(514.02)	(710.43)	(711.19)	(553.65)
p (interaction = 0)	0.28	0.82	0.11	0.77
Panel B. Shares resources				
Audience	1,192.14***	976.65**	1,571.13***	902.34***
	(259.48)	(379.47)	(435.75)	(264.92)
× Shares resources	−720.24**	−318.24	−1,174.43**	−516.66
	(331.85)	(504.87)	(537.84)	(362.68)
p (interaction = 0)	0.03	0.53	0.03	0.16
Panel C. Discusses finance				
Audience	1,139.03***	1,066.49***	1,416.23***	841.97***
	(231.74)	(338.35)	(378.90)	(304.98)
× Discusses finance	−725.96**	−516.15	−1,066.02**	−482.37
	(295.00)	(428.22)	(469.55)	(398.79)
p (interaction = 0)	0.02	0.23	0.03	0.23
Panel D. Close friends				
Audience	1,100.25***	1,446.70***	1,065.99***	799.49***
	(251.54)	(346.07)	(386.81)	(298.37)
× Close friends	−669.14**	−1,189.56***	−469.16	−415.37
	(317.82)	(426.61)	(499.02)	(393.95)
p (interaction = 0)	0.04	0.01	0.35	0.29
Observations	5,514	5,514	5,514	5,514
Fixed Effects	Individual, Round	Individual, Round	Individual, Round	Individual, Round
SE Cluster	Session	Session	Session	Session

Standard errors clustered at the session level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B Appendix: Optimal Repayment

An interior optimum $r^*(s)$ satisfies

$$u'(y(\theta_i) - r^*(s)) = \alpha_i N'(r^*(s) - n^I(\theta_i))\mathbf{1}\{r^*(s) < n^I(\theta_i)\} + s\gamma_i N'(r^*(s) - n^R(\mathbf{x}_i))\mathbf{1}\{r^*(s) < n^R(\mathbf{x}_i)\}.$$

Under concavity, the optimum is $r^*(s) = 0$ if $U_r(0; s, \theta_i, \mathbf{x}_i) \leq 0$ and $r^*(s) = \bar{r}$ if $U_r(\bar{r}; s, \theta_i, \mathbf{x}_i) \geq 0$. Otherwise, $r^*(s) \in (0, \bar{r})$ is the unique solution to the first-order condition above.

The second-order condition is

$$u''(y(\theta_i) - r) + \alpha_i N''(r - n^I(\theta_i))\mathbf{1}\{r < n^I(\theta_i)\} + s\gamma_i N''(r - n^R(\mathbf{x}_i))\mathbf{1}\{r < n^R(\mathbf{x}_i)\} < 0.$$

Optimal Repayment under Specific Functional Forms

To illustrate the model, let

$$u(c) = \sqrt{c}, \qquad N(z) = \kappa \min\{z, 0\} - \tfrac{1}{2}[\min\{z, 0\}]^2, \qquad y(\theta_i) = \theta_i \in \{\theta^G, \theta^N, \theta^B\},$$

To simplify, I set both repayment standards equal to the contractual obligation, $n^I(\theta_i) = n^R(\mathbf{x}_i) = \bar{r}$. For $r < \bar{r}$, utility is,

$$U(r; s, \theta_i) = \sqrt{\theta_i - r} - (\alpha_i + s\gamma_i)\left[\kappa(\bar{r} - r) + \tfrac{1}{2}(\bar{r} - r)^2\right],$$

which is strictly concave, since $U'' = -\frac{1}{4}(\theta_i - r)^{-3/2} - (\alpha_i + s\gamma_i) < 0$, so the optimum is unique.

The optimum $r^*(s)$ is

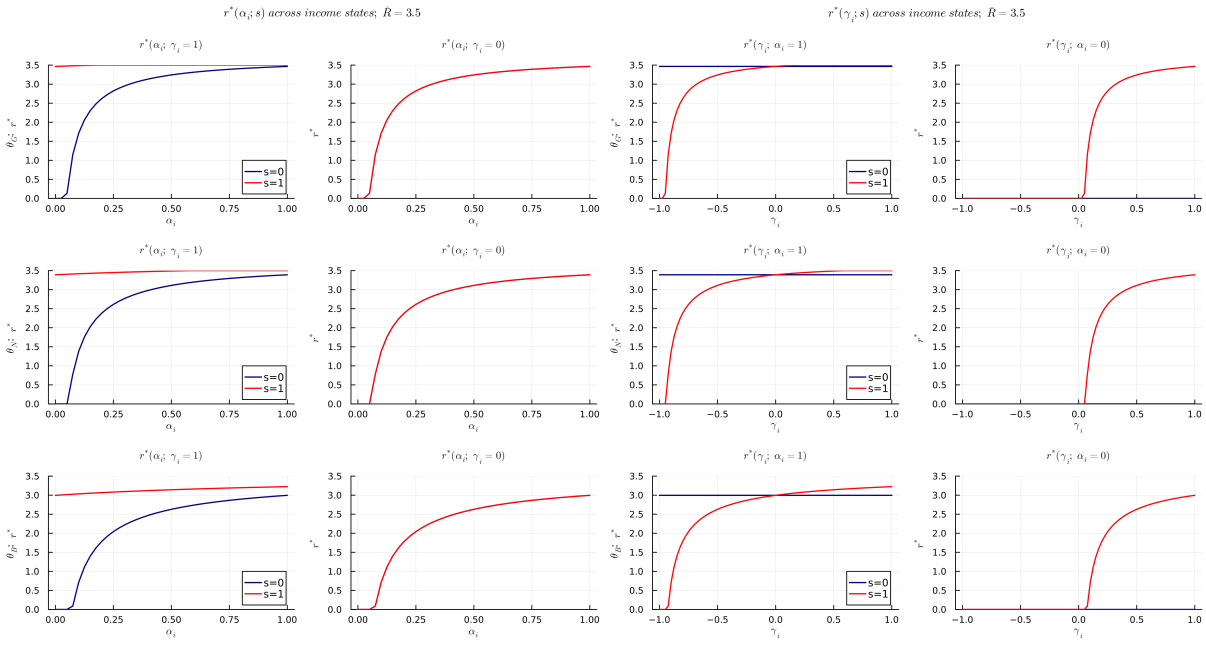
$$r^*(s) \in \begin{cases} 0 & \text{when} \quad (\alpha_i + s\gamma_i)(\kappa + \bar{r}) \leq \dfrac{1}{2\sqrt{\theta_i}}, \\ (0, \bar{r}) & \text{when} \quad \dfrac{1}{2\sqrt{\theta_i - r^*}} = (\alpha_i + s\gamma_i)(\kappa + \bar{r} - r^*), \\ \bar{r} & \text{when} \quad (\alpha_i + s\gamma_i)\kappa \geq \dfrac{1}{2\sqrt{\theta_i - \bar{r}}}. \end{cases}$$

The interior condition is a cubic in r^*, solved numerically. The full repayment occurs as $\kappa > 0$ implies a positive marginal norm penalty at $r = \bar{r}$. Since the two norms enter only through the sum $\alpha_i + s\gamma_i$, which is larger under $s = 1$, and $r^*(s)$ is increasing in that sum, observability weakly raises repayment, $r^*(1) \geq r^*(0)$ for $\gamma_i \geq 0$, with equality possible when both choices remain at the same corner.

Table B.1: Parameterization for the comparative-statics figures

	Value	Description
$\bar{r}$	3.5	Contractual obligation, $r \in [0, \bar{r}]$
$(\theta^G, \theta^N, \theta^B)$	(8.0, 6.0, 3.5)	Good / normal / bad income realization
s	$\{0, 1\}$	Social Observability
α_i	$[0, 1]$	Intrinsic motivation heterogeneity
γ_i	$[-1, 1]$	Reputational motivation heterogeneity
κ	0.2	Penalty slope at benchmark

Figure B.1: Optimal repayment in α_i and γ_i space



C Appendix: Within-Subject Design Diagnostics

C.1 Consistency of Repayment Choices Across States and Rounds

I first assess whether repayment choices exhibit persistent borrower-specific variation within each experimental condition rather than idiosyncratic variation across repeated decisions. Following Castillo et al. (2018), I compare the observed consistency of repayment choices with Monte Carlo distributions generated under random choice.

The design permits two consistency measures. Panel A of Table C.1 treats the good, normal, and bad income states as three items and estimates a separate coefficient for each round and experimental condition. This measure assesses whether borrowers who repay relatively more in one income state also tend to repay relatively more in the other states. Panel B treats analysis rounds two through four as the three items while holding the income state and experimental condition fixed. This measure assesses whether borrowers maintain their relative repayment behavior across repeated decisions.

Each scale contains three items. The standardized Cronbach's alpha reported in the table is

$$\texttt{Cronbach's alpha} = \frac{\rho_{12} + \rho_{13} + \rho_{23}}{1 + \frac{2}{3}\left(\rho_{12} + \rho_{13} + \rho_{23}\right)},$$

where ρ_{12}, ρ_{13}, and ρ_{23} denote the pairwise correlations among the three repayment choices used as items in the corresponding scale. A higher coefficient indicates greater persistence in between-borrower differences across the three items. The coefficient therefore measures consistency in borrowers' relative repayment behavior rather than constancy of repayment amounts.

The coefficients in Panel A range from 0.791 to 0.809, indicating substantial consistency in borrowers' relative repayment choices across income states. The coefficients in Panel B range from 0.810 to 0.869, indicating similarly strong consistency across rounds. The coefficients are similar in magnitude across the private and audience games, with no apparent reduction in consistency under social observability. They also remain similar across rounds despite the decline in average repayment over the session. Across-round consistency is numerically highest in the bad state, with coefficients of 0.849 in the private game and 0.869 in the audience game.

I compare the across-round coefficients in Panel B with two random-choice Monte Carlo distributions. The first is generated by drawing each repayment independently from the eight-point feasible repayment grid. The second is generated by independently resampling repayment from the observed distribution in each round within the corresponding income state and experimental condition. The latter preserves the round-specific marginal distributions of repayment while removing the borrower-specific association across rounds. Under both procedures, the three repayment choices are drawn independently, so their population correlations are zero and any sample correlation arises from finite-sample variation.

The resulting Cronbach's alpha distributions are centered near zero. The 95th percentile is 0.085 under random draws from the uniform repayment grid and ranges from 0.085 to 0.095 under empirical-marginal resampling across the six state-by-condition cells. None of the 1,000 Monte Carlo draws reaches the corresponding observed coefficient, yielding a Monte Carlo p-value of 0.001 in each cell. The observed across-round coefficients are therefore substantially larger than those generated by random choice, including when the observed round-specific repayment distributions are preserved.

These results are consistent with persistent borrower-level heterogeneity in repayment behavior across states and rounds. They do not imply that repayment remains constant in levels over time or that borrower-specific audience effects are stable across rounds. The coefficients measure persistence in repayment levels within each condition, not persistence in the within-borrower difference between the audience and private conditions. Changes in repayment levels and in the average audience effect across rounds are examined separately in Appendix C.2.

Table C.1: Consistency of repayment choices across states and rounds

	Private	*Audience*
Panel A. Across income states		
Round 2	0.807	0.796
Round 3	0.798	0.799
Round 4	0.791	0.809
Panel B. Across analysis rounds		
Good yield	0.841	0.832
Normal yield	0.822	0.810
Bad yield	0.849	0.869
Items per scale	3	3
Borrowers	919	919
Random-choice Monte Carlo comparison for Panel B		
Uniform grid, 95th percentile	0.085	
Empirical marginal, 95th percentile	0.085–0.095	

Notes. Entries report standardized Cronbach's alpha coefficients. Panel A uses the three income states as items, while Panel B uses analysis rounds two through four as items. Private and audience choices are evaluated separately. The uniform-grid comparison draws repayment independently from the eight-point choice set. The empirical-marginal comparison independently resamples from the observed round-specific repayment distributions. Both use 1,000 Monte Carlo draws of 919 borrowers. The empirical-marginal entry reports the range of 95th percentiles across the six state-by-condition cells.

C.2 Pooling Across Rounds

Because each borrower is observed under both experimental conditions in each of three rounds, I examine whether repayment changes systematically over the session and whether the audience effect varies across rounds or depends on treatment order. Following List (2025), this section examines round trends in repayment, variation in the audience effect across rounds, and asymmetry by treatment order. Repayment levels decline over the session, but the audience effect does not vary detectably across rounds and there is no detectable order asymmetry. These patterns support reporting a pooled audience

effect across rounds 2–4 with round fixed effects, while recognizing that the estimates are not sufficiently precise to rule out moderate round-specific variation.

C.2.1 Round Trends in Repayment

Figure C.1 plots mean repayment by round for each income state and experimental condition. Repayment declines from round 2 to round 4 in both the private and audience games.

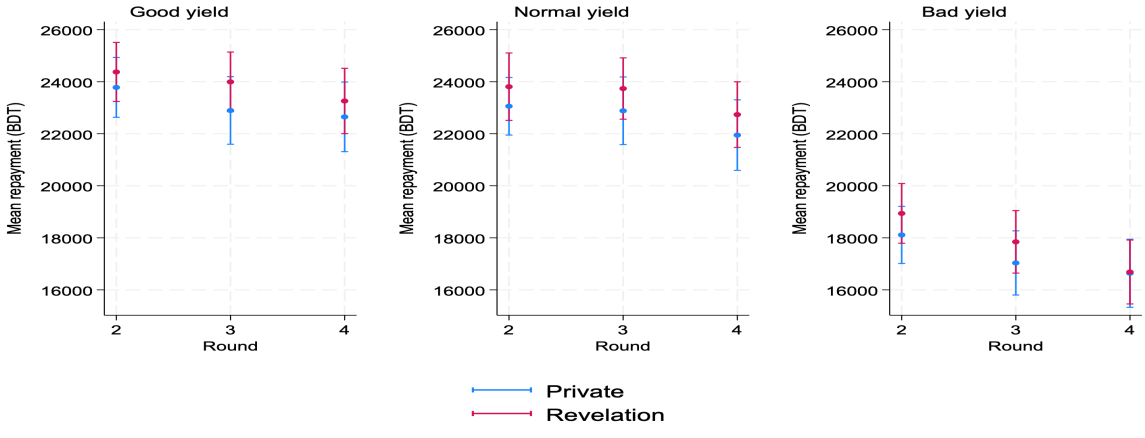
To quantify this pattern, I estimate

$$R_{it} = \mu_i + \sum_{t=3}^{4} \delta_t \mathbb{1}[\text{round} = t] + \beta \, \text{Audience}_{it} + \varepsilon_{it}, \tag{11}$$

separately by income state, where R_{it} denotes borrower i's repayment choice in round t and μ_i is a borrower fixed effect. Standard errors are clustered at the session level. Round 2 and the private condition are the omitted categories.

Panel A of Table C.2 shows that the round coefficients are jointly different from zero in all three income states. Relative to round 2, repayment in round 4 is lower by 1,863 Taka in the bad state, 1,091 Taka in the normal state, and 1,124 Taka in the good state. The decline therefore reflects a general change in repayment levels over the session rather than a change specific to the audience game. Because round is perfectly correlated with position in the session, the data do not distinguish learning, fatigue, or other time-related explanations for this decline.

Figure C.1: Repayment Levels by Round



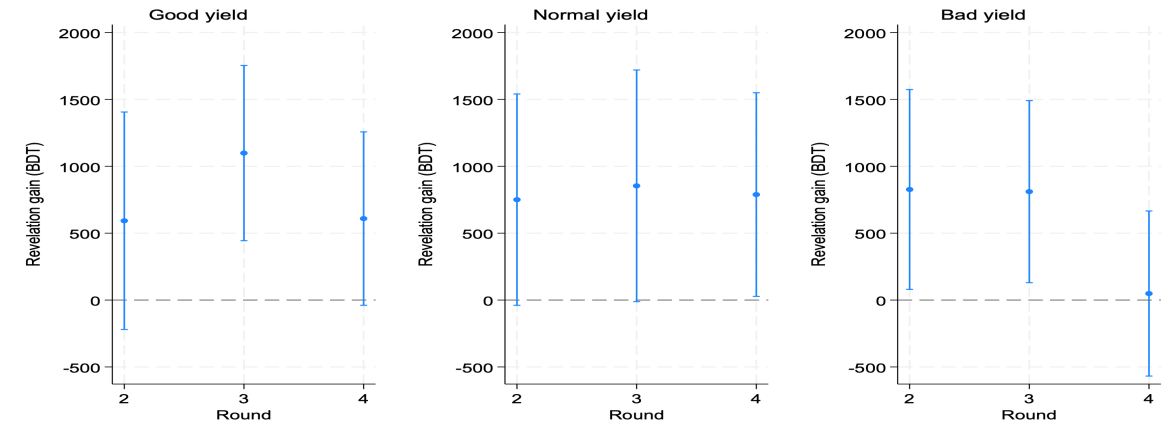
95% CIs, clustered at session level. Rounds 2-4.

C.2.2 Stability of the Audience Effect Across Rounds

I next examine whether the audience effect varies systematically across rounds. Panel B of Table C.2 augments equation (11) with round-by-audience interactions. The interactions are jointly indistinguishable from zero in all three income states, with $p = 0.183$ in the bad state, $p = 0.983$ in the normal state, and $p = 0.445$ in the good state.

Figure C.2 plots the implied audience effect separately by round. The point estimates do not show a systematic pattern across rounds, although they are imprecisely estimated. In particular, the interaction standard errors range from approximately 512 to 581 Taka, relative to pooled audience effects of 562 to 798 Taka. The estimates therefore provide no detectable evidence of systematic variation in the audience effect across rounds, but they cannot rule out economically meaningful round-specific differences, particularly in the bad income state.

Figure C.2: Audience Effect by Round



C.2.3 Order Effects

A separate concern in a within-subject design is that exposure to one experimental condition may affect behavior in the other. Because the order of the private and audience games was randomized at the session level, I test whether the audience effect differs between sessions in which the audience game was presented first and those in which the private game was presented first. Panel C reports these specifications without borrower fixed effects.

Panel C of Table C.2 shows no detectable interaction between the audience condition and treatment order in any income state or in the pooled specification. In the pooled model, the interaction is -170 Taka with $p = 0.554$. The state-specific estimates are also statistically indistinguishable from zero. As with the round interactions, these estimates are not sufficiently precise to exclude moderate order effects, but there is no systematic evidence that the audience effect differs depending on which condition borrowers encounter first.

Taken together, these diagnostics support pooling rounds 2–4 while controlling for round fixed effects. Repayment levels change over the session, but the round-by-audience interactions provide no detectable evidence that the audience effect changes systematically with round. The randomized order comparison similarly provides no detectable evidence that the audience effect differs by treatment order. I therefore include round fixed effects throughout the main specifications and interpret the pooled

coefficient as the average audience effect across rounds 2–4.

Table C.2: Round Stability and Order Effects

	(1) *Good Yield Repayment*	(2) *Normal Yield Repayment*	(3) *Bad Yield Repayment*	(4) *Pooled*
Panel A. Round trends, borrower fixed effects				
Round 3	−633.84**	−122.42	−1,085.42***	
	(302.72)	(308.46)	(309.78)	
Round 4	−1,123.50***	−1,090.86***	−1,863.44***	
	(375.14)	(344.79)	(350.63)	
Audience	767.14***	797.97***	562.21***	
	(197.70)	(248.96)	(198.45)	
F, round effects jointly zero	4.54	7.00	14.15	
p-value	0.013	0.001	0.000	
Panel B. Round by audience, borrower fixed effects				
Round 3 × Audience	505.98	103.37	−16.32	
	(569.44)	(581.31)	(512.17)	
Round 4 × Audience	16.32	38.08	−778.02	
	(573.74)	(580.66)	(515.50)	
F, interactions jointly zero	0.82	0.02	1.73	
p-value	0.445	0.983	0.183	
Panel C. Order effects, no borrower fixed effects				
Audience × Audience first	498.76	−518.32	−488.94	−169.50
	(391.39)	(496.38)	(393.16)	(285.07)
F, interaction zero	1.62	1.09	1.55	0.35
p-value	0.206	0.299	0.217	0.554
Observations	5,514	5,514	5,514	16,542
Borrowers	919	919	919	919
Session clusters	100	100	100	100

Notes. The dependent variable is repayment in Taka against an obligation of 35,000 Taka. Standard errors in parentheses are clustered at the session level. Round 2 and the private condition are the omitted categories. Panels A and B include borrower fixed effects and are estimated separately by income state. Panel C tests whether the audience effect differs when the audience game is presented first. Panel C is estimated without borrower fixed effects. The pooled specification includes income-state indicators. Stars denote $^{***}p < 0.01$, $^{**}p < 0.05$, and $^{*}p < 0.1$.

C.3 Precision of Estimates

The stacked specifications provide no detectable evidence that the audience effect differs across income states. Because failure to reject differences does not establish equality, particularly when the interaction terms are imprecisely estimated, I summarize the precision of these estimates using approximate minimum detectable effects. I report these rather than observed power, which is a monotone transformation of the realized p-value and provides little additional information once the estimate is observed (Hoenig and Heisey, 2001). Table C.3 reports approximate minimum detectable effects for the coefficients of interest, calculated as

$$\text{MDE} = (z_{0.975} + z_{0.80})\,\widehat{\text{se}},$$

for a two-sided test at the 5 percent level with 80 percent power.

Panel A reports the fixed effects estimates on observed repayment. The audience effect is estimated

more precisely than the state interactions. Its minimum detectable effect is 554 Taka, or approximately 2.4 percent of private-game repayment in the good income state, compared with an estimated effect of 767 Taka. The state interactions are less precisely estimated. The minimum detectable difference is approximately 823 Taka for the normal-state interaction and 721 Taka for the bad-state interaction.

The 95 percent confidence interval for the bad-state interaction ranges from −716 to 306 Taka. The estimate therefore remains compatible with differences of several hundred Taka across income states. The estimates provide no detectable evidence of state heterogeneity in the audience effect, but they do not establish that the effect is identical across states.

Panel B reports the corresponding precision for the two-limit Tobit estimates of latent repayment. The same pattern holds. The audience coefficient is estimated more precisely than the state interactions, while the confidence intervals around the interaction terms remain wide. Because the Tobit coefficients refer to latent repayment, their levels and minimum detectable effects are not directly comparable with the fixed effects estimates on observed repayment.

The round-by-audience interactions in Section C.2 are less precise still. Using the same calculation, their minimum detectable effects range from approximately 1,435 to 1,629 Taka, compared with pooled audience effects of 562 to 798 Taka. The data therefore cannot establish that the audience effect is constant across rounds. I instead interpret the pooled specification as estimating the average audience effect across rounds 2–4 while recognizing that moderate round-specific variation cannot be ruled out.

Table C.3: Precision of the Audience and State-Interaction Estimates

	Estimate	S.E.	MDE	95% CI
Panel A. OLS fixed effects				
Audience	767.1	197.7	553.9	—
Normal × Audience	30.8	293.8	823.1	[−552, 614]
Bad × Audience	−204.9	257.3	721.0	[−716, 306]
Panel B. Two-limit Tobit				
Audience	1,496.5	379.9	1,064.4	—
Normal × Audience	−239.2	549.4	1,539.2	[−1,316, 838]
Bad × Audience	−676.5	450.0	1,260.6	[−1,558, 205]

Notes. The minimum detectable effect is approximated by $(z_{0.975} + z_{0.80}) \times \widehat{\text{se}}$ and gives the approximate coefficient magnitude associated with 80 percent power in a two-sided test at the 5 percent level, conditional on the estimated standard error. Standard errors are clustered at the session level. Estimates, standard errors, and confidence intervals are taken from Table 5.3. The Tobit estimates refer to latent repayment and are not directly comparable with the OLS estimates on observed repayment.

D Appendix: Monte Carlo Assessment of Censored Outcome Estimators

I use Monte Carlo simulations to assess the interpretation of the fixed effects and two-limit Tobit estimates when repayment is censored at the experimental bounds. The exercise distinguishes the audience effect on observed repayment, $\beta^O = E[y_{ir}^A - y_{ir}^P]$, from the audience effect on latent repayment, β. I calibrate the simulations to the Normal-state repayment distribution and vary the importance of persistent borrower heterogeneity and dependence in the decision-specific errors.

Data generating process. Suppressing the income state indicator for simplicity, latent repayment in the Private and Audience games is generated as

$$\begin{aligned} y_{ir}^{P*} &= \mu_i + \varepsilon_{ir}^P, \\ y_{ir}^{A*} &= \mu_i + \beta + \varepsilon_{ir}^A, \end{aligned} \tag{12}$$

where μ_i is the individual specific latent repayment level and β is the audience effect on latent repayment. Each simulated sample contains 919 individuals observed in three rounds under both experimental conditions, giving 5,514 observations. I set $\beta = 1{,}000$ Taka throughout the simulation so that differences across simulation cells reflect differences in the data generating process rather than changes in the underlying audience effect. Observed repayment is censored at the experimental bounds, $y_{ir}^g = \min\{\max(y_{ir}^{g*}, 0), \bar{Y}\}$, where $\bar{Y} = 35{,}000$ Taka and $g \in \{P, A\}$.

Both experimental conditions have total latent repayment variance $\text{Var}(y_{ir}^{P*}) = \text{Var}(y_{ir}^{A*}) = \sigma^2 = \sigma_\mu^2 + \sigma_\varepsilon^2$, and $\text{Var}(\varepsilon_{ir}^P) = \text{Var}(\varepsilon_{ir}^A) = \sigma_\varepsilon^2$.

For a given σ^2, let ϕ_μ denote the share of total latent repayment variance attributable to individual heterogeneity.

$$\phi_\mu \equiv \frac{\sigma_\mu^2}{\sigma^2}, \qquad \sigma_\mu^2 = \phi_\mu \sigma^2, \qquad \sigma_\varepsilon^2 = (1 - \phi_\mu)\sigma^2.$$

Individual specific repayment levels are distributed as

$$\mu_i \overset{\text{iid}}{\sim} N\left(\bar{\mu}, \sigma_\mu^2\right),$$

independently of the decision-specific errors.

I allow the decision specific errors to have an exchangeable correlation structure across rounds. For each individual i, let $\boldsymbol{\varepsilon}_i = (\varepsilon_{i1}^P, \varepsilon_{i1}^A, \ldots, \varepsilon_{i3}^P, \varepsilon_{i3}^A)^\top$. The error vectors are independently and identically distributed across individuals according to

$$\boldsymbol{\varepsilon}_i \sim N\left(\mathbf{0}, \sigma_\varepsilon^2 \boldsymbol{\Sigma}\right),$$

where the (r, r') block of $\boldsymbol{\Sigma}$ is

$$\boldsymbol{\Sigma}_{rr'} = \begin{cases} \begin{pmatrix} 1 & \rho_\varepsilon \\ \rho_\varepsilon & 1 \end{pmatrix} & \text{if } r = r', \\ \begin{pmatrix} \rho_{\text{round}} & \rho_{\text{round}}\,\rho_\varepsilon \\ \rho_{\text{round}}\,\rho_\varepsilon & \rho_{\text{round}} \end{pmatrix} & \text{if } r \neq r'. \end{cases}$$

The parameter ρ_ε determines the within-round cross-game correlation in idiosyncratic deviations, while ρ_{round} governs the cross-round correlation in idiosyncratic deviations within the same game. Thus, $\text{Cor}(\varepsilon^g_{ir}, \varepsilon^g_{ir'}) = \rho_{\text{round}}$ for $g \in \{P, A\}$ and $r \neq r'$, while $\text{Cor}(\varepsilon^P_{ir}, \varepsilon^A_{ir'}) = \rho_\varepsilon \rho_{\text{round}}$ for $r \neq r'$. Under this separable correlation structure, the cross-round cross-game correlation is the product of the two parameters. Setting $\rho_{\text{round}} = 0$ allows independence across rounds while retaining the within-round cross-game correlation ρ_ε, as

$$\begin{pmatrix} \varepsilon^P_{ir} \\ \varepsilon^A_{ir} \end{pmatrix} \overset{\text{iid}}{\sim} N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix},\ \sigma^2_\varepsilon \begin{pmatrix} 1 & \rho_\varepsilon \\ \rho_\varepsilon & 1 \end{pmatrix} \right].$$

Estimators and estimands. I compare three estimators. First, the linear fixed effects specification is

$$y_{igr} = \nu_i + \beta^F Audience_{igr} + u_{igr},$$

where β^F estimates the observed audience effect β^O.
The two-limit Tobit specification is

$$y^*_{igr} = \alpha + \beta^T Audience_{igr} + u_{igr}, \qquad y_{igr} = \min\{\max(y^*_{igr}, 0), \bar{Y}\},$$

where β^T estimates the latent audience effect β.

The audience effect is subject to an additional layer of censoring beyond the bounds on repayment itself. Because repayment is bounded between zero and $\bar{Y}$, the observed difference between Audience and Private repayment is also bounded, with the feasible range depending on the borrower's Private-game repayment. In particular, borrowers with high Private repayment have less room to increase observed repayment in the Audience game. This feature makes a variable-limit Tobit a natural alternative for estimating the latent audience effect, since the censoring limits of the repayment difference vary across borrowers. I therefore include this estimator in the Monte Carlo exercise to evaluate its performance, although I do not use it in the main analysis. The variable-limit Tobit defines $\widetilde{y}_{ir} = y^A_{ir} - \bar{y}^P_i$, where $\bar{y}^P_i = \frac{1}{3}\sum_r y^P_{ir}$, and estimates

$$\widetilde{y}^*_{ir} = \beta^V + u_{ir}, \qquad \widetilde{y}_{ir} = \min\{\max(\widetilde{y}^*_{ir}, -\bar{y}^P_i), \bar{Y} - \bar{y}^P_i\},$$

where β^V is interpreted as the latent audience effect relative to the observed Private-game repayment benchmark.

Calibration and parameters. I calibrate $\bar{\mu}$ and σ to the pooled lower- and upper-censoring shares in the Normal state. Because the sample contains equal numbers of Private and Audience observations, the two latent distributions, $N(\bar{\mu}, \sigma^2)$ and $N(\bar{\mu} + \beta, \sigma^2)$, receive equal weight. Matching the empirical shares at zero and 35,000 gives $\bar{\mu} = 26{,}625$ and $\sigma = 21{,}668$. I then vary ϕ_μ, ρ_ε, and ρ_{round}, using 500 replications per cell. Each simulated sample contains 919 borrowers observed in three rounds under both conditions, giving 5,514 observations.

Table D.1: Monte Carlo Parameterization

Parameter	Symbol	Value	Role
Borrowers	n	919	Simulated estimation sample
Rounds	r	3	Repeated decisions per individual in each game
True latent audience effect	β	1,000	Target latent estimand
Mean latent repayment level	$\bar{\mu}$	26,625	Calibrated mean of latent repayment
Total standard deviation	σ	21,668	Calibrated dispersion of latent repayment
Individual variance share	ϕ_μ	0.30, 0.50, 0.70, 0.90	Share of latent variance from individual heterogeneity
Cross-game error correlation	ρ_ε	-0.9, -0.6, -0.3, 0.00, 0.3, 0.6, 0.9	Correlation of Private and Audience errors within a round
Cross-round error correlation	ρ_{round}	0.00, 0.3, 0.6, 0.9	Correlation across rounds
Experimental bounds	$[0, \bar{Y}]$	[0, 35,000]	Lower and upper repayment limits
Replications	B	500	Number of Monte Carlo replications

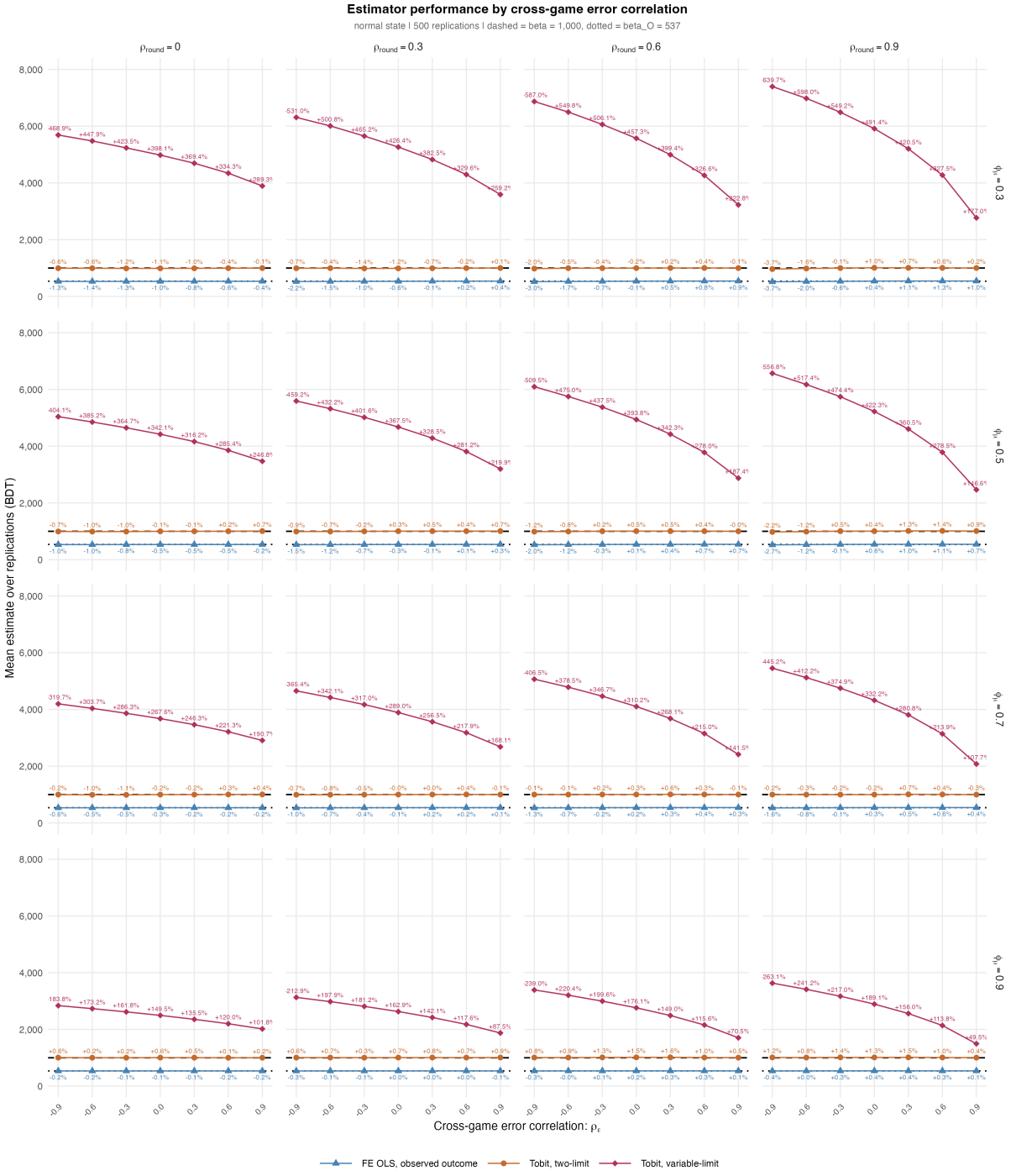
Results. Figure D.1 reports the simulation results. The fixed effects estimator remains centered around the observed effect, $\beta^O = 537$, with deviations between -3.8 and $+1.5$ percent across the simulation grid. The two-limit Tobit similarly remains centered around the latent effect, with deviations between -3.7 and $+1.6$ percent of β. The correlation parameters alter dependence across repeated observations but leave the marginal latent distributions unchanged, and neither point estimate is materially affected by this dependence.

The two estimators therefore recover different quantities. Under the Normal-state calibration, $\beta^O/\beta = 0.537$, so censoring reduces the observed repayment response to a little over half of the latent response. For a small shift in latent repayment, this ratio is approximately the uncensored share of the repayment distribution, which is also 0.537 in the calibration. The fixed effects and Tobit estimates should therefore be interpreted as the audience effects on observed and latent repayment, respectively, rather than as competing estimates of the same parameter.

The variable-limit Tobit, by contrast, substantially overstates β. The observed Private benchmark $\bar{y}_i^P$ is a censored and noisy proxy for μ_i, so the latent transformed outcome is $y_{ir}^{A*} - \bar{y}_i^P = \beta + (\mu_i - \bar{y}_i^P) + \varepsilon_{ir}^A$ rather than β plus a mean-zero error. I therefore do not use this specification in the main analysis.

The simulation establishes that, under the maintained censored-normal data generating process, the fixed effects and two-limit Tobit estimators recover their intended observed and latent audience effects across a wide range of persistent borrower heterogeneity and within-borrower error dependence.

Figure D.1: Monte Carlo Performance of Censored Outcome Estimators



Notes. The figure reports mean estimates across 500 Monte Carlo replications under the Normal-state calibration. The dashed line denotes the true latent audience effect, $\beta = 1{,}000$, and the dotted line denotes the implied observed audience effect, $\beta^{O} = 537$. Rows vary the individual variance share ϕ_{μ}, columns vary the cross-round error correlation ρ_{round}, and the horizontal axis varies the within-round cross-game error correlation ρ_{ε}. Percentage labels report bias relative to the corresponding target.